PEARSON EDEXCEL INTERNATIONAL A LEVEL PHYSICS · WPH15

Unit 5: Thermodynamics, Radiation, Oscillations and Cosmology

Revision notes, spec point by spec point (Thermodynamics; Nuclear Decay; Oscillations; Astrophysics and Cosmology)

How to use these notes. Every section starts with a spec-point badge and a short summary of what that spec point asks, so you always know which point you are revising. The header at the top of each page names the spec point(s) on that page.

BEYOND THE SPEC, BUT EXAMINED

Content the specification does not spell out, but that past-paper mark schemes have rewarded.

EXAM TIP

How to pick up the marks.

MARK-SCHEME LANGUAGE

Words and phrases mark schemes credit.

COMMON MISTAKE

What examiners keep seeing go wrong.

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Topic 5.3 – Thermodynamics

KEY IDEA

How Unit 5 calculations are marked. Every calculation earns "use of" marks for each step you write down, so show each substitution. Give a final answer with its **unit**. In a **"show that"** question, give your answer to **one more significant figure** than the value printed (3 s.f. for a 2 s.f. value) and never just quote it back. When the question makes a **claim**, the last mark is a **numerical comparison plus a verdict** ("48 °C is less than 60 °C, so the claim is not correct"). Temperatures in gas, kinetic-theory and black-body equations must be in **kelvin**: T/K = θ/°C + 273.
The data and formulae booklet gives every equation in this unit, plus k, G, σ, c, e and u. It does **not** give $4\pi r^2$ (area of a sphere), $\frac{4}{3}\pi r^3$ (volume of a sphere), ρ = m/V, p = F/A or the centripetal force equations: learn those.

SPEC 125 Specific heat capacity and specific latent heat

WHAT THIS SPEC POINT ASKS

Use ΔE = mcΔθ for a temperature change and ΔE = LΔm for a change of state.

125 Specific heat capacity

- The **specific heat capacity** c of a material is the energy needed to raise the temperature of **1 kg** of it by **1 K** (or 1 °C).

$$\Delta E = mc\Delta\theta$$

- ΔE = energy transferred (J); m = mass (kg); c = specific heat capacity (J kg^{-1} K^{-1}); Δθ = temperature **change** (K or °C).
- A kelvin and a degree Celsius are the same size, so a temperature **difference** is the same number in both. Don't add 273 to a Δθ.
- Water has a high c (about 4200 J kg^{-1} K^{-1}): it takes a lot of energy to warm, so it is a good coolant and heat store. Metals have low c (copper about 390 J kg^{-1} K^{-1}): they heat up quickly. You are always given c in the question.

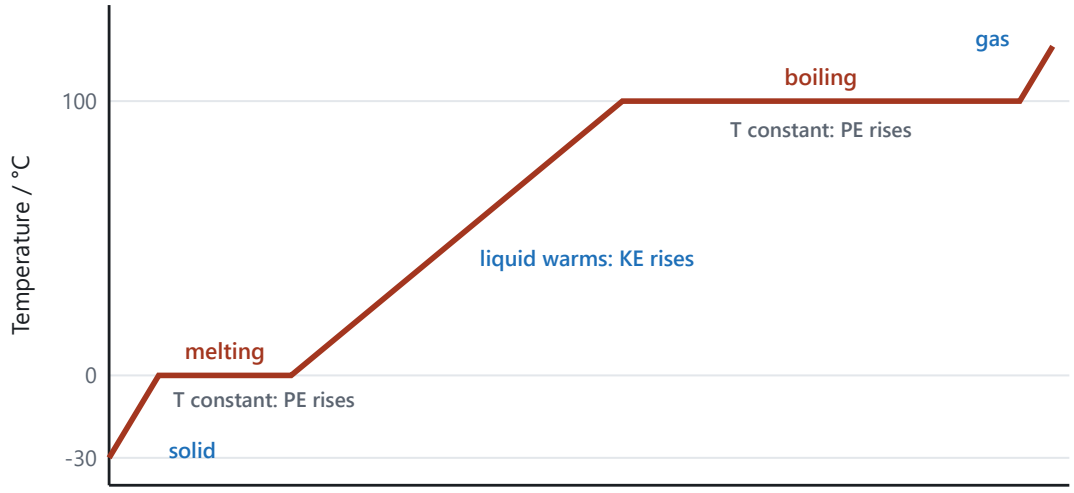
125 Specific latent heat

- During a **change of state** (melting, boiling, freezing, condensing) energy is transferred but the **temperature stays constant**.
- The **specific latent heat** L is the energy needed to change the state of **1 kg** of a substance **without changing its temperature**.
  - **Fusion** (L_f): solid ↔ liquid. Water: 3.34 × 10^5 J kg^{-1}.
  - **Vaporisation** (L_v): liquid ↔ gas. Water: 2.26 × 10^6 J kg^{-1}.

$$\Delta E = L\Delta m$$

- ΔE = energy (J); L = specific latent heat (J kg^{-1}); Δm = mass that changes state (kg).
- L_v is about 7 times L_f. Melting only loosens the molecules (they stay close together). Boiling separates them completely, and the gas also has to push back the atmosphere as it expands, so far more energy is needed per kilogram.

125 The heating curve



Heating ice at a constant power (sketch, not to scale). On the sloping parts the energy raises molecular kinetic energy; on the flat parts it raises molecular potential energy. The boiling plateau is really about 7 times as long as the melting one, because $L_v \approx 7L_f$.

- **Sloping sections**: one state is warming; $\Delta E = mc\Delta\theta$. The liquid line is less steep than the solid line because water's c (4200) is about twice ice's c (2100 J kg^{-1} K^{-1}).
- **Flat sections**: state is changing; $\Delta E = L\Delta m$. Temperature (so mean molecular KE) is constant; the energy goes into **potential energy** of the molecules as bonds between them are broken.
- At constant power P, the time on each section is $\Delta E/P$. So a flat section's length is proportional to L, and a sloping section's gradient is $P/(mc)$.

125 Linking to power and time

$$\Delta E = Pt \quad \text{or} \quad P = \Delta E/t$$

- An electric heater: $P = VI$, so $\Delta E = VIt$. For a flowing liquid, use the mass per second: $P = (m/t)c\Delta\theta$.

WORKED EXAMPLE: HEAT THEN BOIL

A 2.4 kW kettle holds 1.5 kg of water at 18 °C. It heats the water to 100 °C and then boils away 0.20 kg. Assume all the energy goes into the water. c = 4180 J kg^{-1} K^{-1}, $L_v = 2.26 \times 10^6$ J kg^{-1}. How long is the kettle on?

Heating: $\Delta E_1 = mc\Delta\theta = 1.5 \times 4180 \times (100 - 18) = 5.14 \times 10^5$ J

Boiling: $\Delta E_2 = L\Delta m = 2.26 \times 10^6 \times 0.20 = 4.52 \times 10^5$ J

Total: $\Delta E = 5.14 \times 10^5 + 4.52 \times 10^5 = 9.66 \times 10^5$ J

$t = \Delta E/P = 9.66 \times 10^5 / 2400 = 403$ s, so **t ≈ 400 s (2 s.f.)**.

A real kettle takes longer: energy is also transferred to the kettle and the surroundings.

WORKED EXAMPLE: ICE IN A DRINK (ENERGY BALANCE)

0.030 kg of ice at 0 °C is put into 0.25 kg of water at 20 °C. Find the final temperature θ. Ignore the cup and the surroundings. $L_f = 3.34 \times 10^5$ J kg^{-1}, c = 4180 J kg^{-1} K^{-1}.

Energy lost by the warm water = energy to melt the ice + energy to warm the melted ice from 0 °C to θ:

$0.25 \times 4180 \times (20 - \theta) = 0.030 \times 3.34 \times 10^5 + 0.030 \times 4180 \times \theta$

$20\,900 - 1045\theta = 10\,020 + 125.4\theta$

$\theta = 10\,880 / 1170.4 =$ **9.3 °C**

EXAM TIP

Lay every multi-step question out the way the mark scheme splits it: (1) mass from ρ = m/V if you're given a volume, (2) mcΔθ, (3) LΔm, (4) add them, (5) P = ΔE/t, (6) compare with the claim. Each step is a mark. Pick the right L: **fusion** for melting or freezing, **vaporisation** for boiling, evaporating or condensing.

MARK-SCHEME LANGUAGE

- "Use of ΔE = mcΔθ", "use of ΔE = LΔm", "use of P = ΔE/t", "both energies added".
- "Substitution for Δθ must be a temperature difference. Ignore attempt to convert to kelvin" (WPH15 Jan 2026).
- Why the real value differs: "energy is transferred (from the water) to the surroundings" / "not all of the energy from the heater is used to raise the water temperature".
- In a flat section: "temperature (so kinetic energy) is constant", "the energy increases the **potential energy** of the molecules".

Often asked as: kettle or pool heating then boiling; melting ice (ice palace, ice cubes in a drink, ice in fuel); coolant flow rate; reading c or L from a graph; comparing a calculated energy, time or mass with a claim.

COMMON MISTAKE

Forgetting one of the two energies (the latent heat part is the one most often lost). Writing E = P/t instead of E = Pt. Using the wrong c when two substances are present. Mixing up energy and power. Using L_f for boiling. Not finishing with the comparison the question asked for. In a "show that", giving the same number of s.f. as the printed value.

BEYOND THE SPEC, BUT EXAMINED

- **Efficiency** = useful energy (or power) out ÷ total in × 100%: e.g. a microwave oven's efficiency from the temperature rise it produces (WPH15 Oct 2020).
- Reading a latent heat from the gradient of an energy-against-mass graph (WPH15 Jan 2026).
- When gravitational energy becomes thermal energy (a falling droplet), mgΔh = mcΔθ, so **mass cancels** and Δθ doesn't depend on the mass (WPH15 Jun 2022).
- A lid on a pan saves energy: less steam escapes, and steam condensing on the lid gives its latent heat back to the pan (WPH15 Jan 2025).

SPEC 126 Core practical 12: thermistor in a potential divider as a thermostat

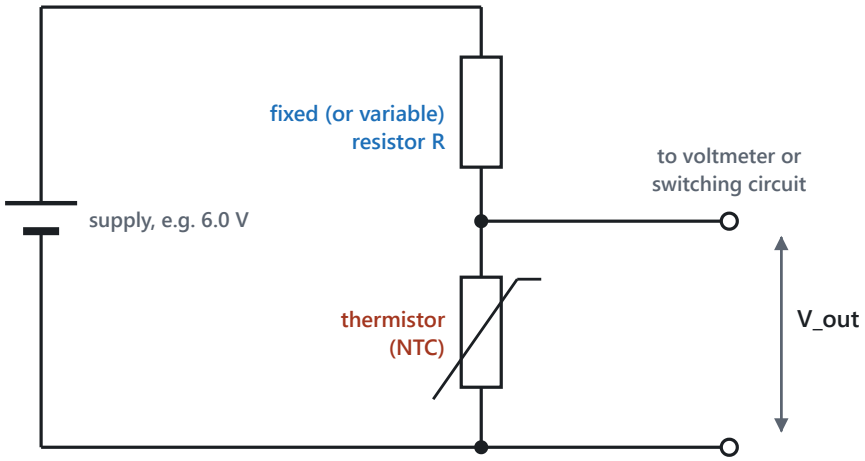
WHAT THIS SPEC POINT ASKS

Calibrate a thermistor in a potential divider so that its output voltage can be used to measure, or switch on at, a set temperature.

126 The physics you need

- An **NTC thermistor**: its resistance **falls** as temperature rises (more charge carriers are released). The R–θ graph is a curve, steep at low temperature.
- In a **potential divider** of a fixed resistor R and the thermistor R_T across a supply V_{in}, the output across the thermistor is

$$V_{out} = V_{in} \times R_T / (R + R_T)$$



Thermistor potential divider. As the thermistor warms, R_T falls, so V_{out} across it falls and the p.d. across R rises.

- **Thermostat idea**: the output feeds a switch (a transistor or relay) that turns on at a set voltage. Take the output across the **fixed resistor** to switch a cooling fan on when it gets **hot**; take it across the **thermistor** to switch a heater on when it gets **cold**. Making R a **variable resistor** lets you set the switching temperature.

CORE PRACTICAL 12: METHOD

1. Connect the thermistor to an ohmmeter (or a supply, ammeter and voltmeter, with R = V/I) for the calibration of R against θ. For the thermostat, connect it in the potential divider with a voltmeter across the output.
2. Put the thermistor in a beaker of water with a thermometer (or temperature sensor and data logger) next to it. Add **ice** to start at about 0 °C.
3. **Heat** the water gently (Bunsen burner, or a kettle/immersion heater) towards 100 °C, **stirring** so the temperature is the same throughout.
4. Every 5 °C or so, stir, wait for the reading to settle, then read the temperature and the resistance (or V_{out}) at the same moment.
5. Plot **R against θ** (or V_{out} against θ): this is the **calibration curve**. To use it as a thermometer, measure R or V_{out} and read off θ.
6. Repeat as the water cools and average the two readings at each temperature.

Variables: independent = temperature; dependent = resistance (or V_{out}); control = supply p.d., fixed resistor value, same thermistor.
Safety: hot water and a hot tripod; keep leads away from the flame; keep electrical meters away from spilled water.

WORKED EXAMPLE: OUTPUT VOLTAGE

A 6.0 V supply, a 1.5 kΩ fixed resistor and a thermistor whose resistance is 1.0 kΩ at 40 °C. Find V_{out} across the thermistor at 40 °C.
$V_{out} = 6.0 \times 1.0/(1.5 + 1.0) =$ **2.4 V**
If the thermistor warms to 60 °C its resistance falls, so V_{out} across it falls below 2.4 V.

126 Sources of error and improvements

- The thermometer and thermistor must be at the **same temperature**: put them side by side and stir.
- Small current only: a large current **heats the thermistor itself**, so switch off between readings if you use V and I.
- The thermometer's resolution (1 °C) limits precision; use a digital sensor with a data logger for many readings.
- Zero the ohmmeter; read the thermometer at eye level.

MARK-SCHEME LANGUAGE

Indicative content from the 6-mark planning questions (WPH15 Oct 2023, Oct 2024):

- thermistor connected to an ohmmeter, or to an ammeter and voltmeter;
- placed in a water bath / beaker of water; ice added to reach 0 °C, then heated to near 100 °C;
- "use a thermometer to **measure the temperature**" (tie the reading to the resistance "for a particular temperature");
- stir so the temperature is uniform;
- plot resistance against temperature to give the calibration curve; repeat or take enough readings across the range.

Often asked as: a 6-mark "describe how to calibrate" question with a labelled diagram. The potential-divider (thermostat) part has not yet been examined in WPH15, but it uses Unit 4 potential-divider physics.

COMMON MISTAKE

Writing a list of apparatus instead of a method in order. Saying "use a thermometer" without saying it measures the temperature. Confusing the thermistor with a thermometer, or saying the ohmmeter measures p.d. Writing the whole answer about an ordinary resistor. Giving no realistic way to heat or cool the water.

BEYOND THE SPEC, BUT EXAMINED

- The range matters: water lets you go from 0 °C (melting ice) to about 100 °C (boiling), so the calibration covers that range only (WPH15 Oct 2023).

SPEC 127 Core practical 13: specific latent heat

WHAT THIS SPEC POINT ASKS

Measure the specific latent heat of a change of state (fusion of ice, or vaporisation of water).

127 Principle

Supply a **known amount of energy** with an electrical heater, measure the **mass that changes state**, then

$$L = \Delta E/\Delta m = VIt/\Delta m$$

CORE PRACTICAL 13: METHOD (FUSION OF ICE)

1. Set up **two funnels** of crushed ice draining into beakers. One has an immersion heater (the **test**); the other is identical with no heater switched on (the **control**).
2. Wait until the ice is at 0 °C and dripping steadily, then empty both beakers.

3. Switch on the heater for a measured time t. Record V and I (or read a **joulemeter**) to get ΔE = VIt.
4. Weigh the water collected in each beaker. Mass melted by the heater: $\mathbf{\Delta m = m_{test} - m_{control}}$. The control corrects for ice melted by heat from the room.
5. L_f = ΔE/Δm. Repeat and average.

Vaporisation version: stand a beaker of water with an immersion heater on a balance. Once it boils steadily, start timing and record the mass lost in time t. L_v = VIt/Δm. Doing this at two different powers and subtracting cancels the heat lost to the surroundings.
Safety: hot heater element; keep water off the balance and mains; handle ice with tongs.

WORKED EXAMPLE: LATENT HEAT OF FUSION

Heater: 12.0 V, 4.2 A, switched on for 300 s. Water collected: test funnel 52.0 g, control funnel 7.5 g.
ΔE = VIt = 12.0 × 4.2 × 300 = 15 120 J
Δm = 52.0 − 7.5 = 44.5 g = 0.0445 kg
L_f = 15 120 / 0.0445 = $\mathbf{3.4 \times 10^5\ J\ kg^{-1}}$ (accepted value 3.34×10^5 J kg^{-1})

127 Errors: which way do they push L?

Error	Effect on the calculated L
Energy from the heater lost to the surroundings (ΔE used is more than the energy reaching the ice)	L too **high**
Ice melted by the room counted as melted by the heater (no control)	L too **low**
Heater left on longer than the recorded time	measured Δm too big for the ΔE recorded, so L too **low**
Ice not at 0 °C (some energy warms it first)	L too **high**

MARK-SCHEME LANGUAGE

- L = (energy transferred)/(mass changed); with a heater, "L = Pt/Δm".
- "Energy escapes to the surroundings, so the calculated L is too high."
- A control funnel "accounts for ice melted by the surroundings".

Often asked as: an MCQ on which error gives a value that is too low or too high (WPH15 Jun 2026); latent-heat calculations in context. The full method has not been asked step by step.

COMMON MISTAKE

Forgetting to convert g to kg. Using the total mass of ice instead of the mass that melted. Thinking a temperature is the dependent variable: it stays at 0 °C; you measure mass.

SPEC 128 Internal energy

WHAT THIS SPEC POINT ASKS

Understand internal energy as the sum of the randomly distributed kinetic and potential energies of the molecules in a system.

128 What it is

- The **internal energy** of a system is the **sum of the kinetic energies and potential energies of all its molecules**, which are **randomly distributed** among them.
  - **Kinetic energy** comes from the random motion of the molecules (vibrating in a solid; moving around in a liquid and gas). Mean KE depends on **temperature**.
  - **Potential energy** comes from the forces between molecules and depends on their **separation**. Breaking the attractions (melting, boiling) increases the PE.
- "Randomly distributed": at any moment some molecules are fast and some slow; the energy is shared out unevenly and keeps changing in collisions. Only the **mean** KE is fixed by temperature.
- Internal energy is **not** the KE of the object as a whole. A ball thrown across a room doesn't get hotter because it is moving.

128 How internal energy changes

Process	KE of molecules	PE of molecules	Internal energy
Heating a solid, liquid or gas (temperature rising)	rises	(small change)	rises
Melting or boiling at constant temperature	constant	**rises**	rises
Freezing or condensing at constant temperature	constant	**falls**	falls
Cooling (temperature falling)	falls	(small change)	falls

- Internal energy rises when energy is **transferred in by heating** or when **work is done on** the system (e.g. compressing a gas).
- For an **ideal gas** there are no forces between molecules (except in collisions), so there is **no potential energy**. Its internal energy is all kinetic: $U = N \times \frac{3}{2}kT$ (see 132).

WORKED EXAMPLE: INTERNAL ENERGY OF AN IDEAL GAS

A container holds 1.20×10^{24} molecules of an ideal gas at 20 °C. Find its internal energy. $k = 1.38 \times 10^{-23}$ J K^{-1}.

$T = 20 + 273 = 293$ K

Mean KE per molecule $= \frac{3}{2}kT = 1.5 \times 1.38 \times 10^{-23} \times 293 = 6.07 \times 10^{-21}$ J

$U = N \times \frac{3}{2}kT = 1.20 \times 10^{24} \times 6.07 \times 10^{-21} =$ **7.3×10^{3} J**

MARK-SCHEME LANGUAGE

6-mark cooling-curve question (wax cooling, with a flat section XY) (WPH15 Jun 2023):

- "internal energy decreases" as the wax cools;
- "internal energy is the sum of the (random) potential and kinetic energies of the molecules";
- as the temperature falls, "the kinetic energy of the molecules decreases";
- between X and Y the wax is solidifying: "temperature (so KE) is constant" and "the **potential energy** of the molecules decreases".
- Ideal gas: "do not credit NkT as internal energy"; $U = N \times \frac{3}{2}kT$ (WPH15 Oct 2024).

Often asked as: describe how internal energy changes on a heating or cooling curve; calculate U for an ideal gas; MCQ on what mean KE depends on.

COMMON MISTAKE

Saying the PE rises as KE falls "to keep the internal energy constant". Internal energy is not constant: it falls on cooling. Saying internal energy stays the same on a flat section of a cooling curve (the PE is changing). Writing about "energy" or "particles" with no mention of **molecules**. Using $\frac{3}{2}kT$ (one molecule) as the internal energy of the whole gas, or multiplying by N when only one molecule's mean KE was asked for.

BEYOND THE SPEC, BUT EXAMINED

- **$U = \frac{3}{2}pV$** for an ideal gas: $U = N \times \frac{3}{2}kT$ and $NkT = pV$. Derive it by writing these two steps; "do not credit substituting N = 1" (WPH15 Jun 2026).
- Microwave heating: water molecules are made to rotate and collide with neighbours, increasing their KE, so "the internal energy and hence the temperature" rise. Molecules in ice can't rotate, so ice heats poorly (WPH15 Oct 2020).

SPEC 129

Absolute zero and the meaning of temperature

WHAT THIS SPEC POINT ASKS

Understand absolute zero, and that the average kinetic energy of the molecules is proportional to the absolute (kelvin) temperature.

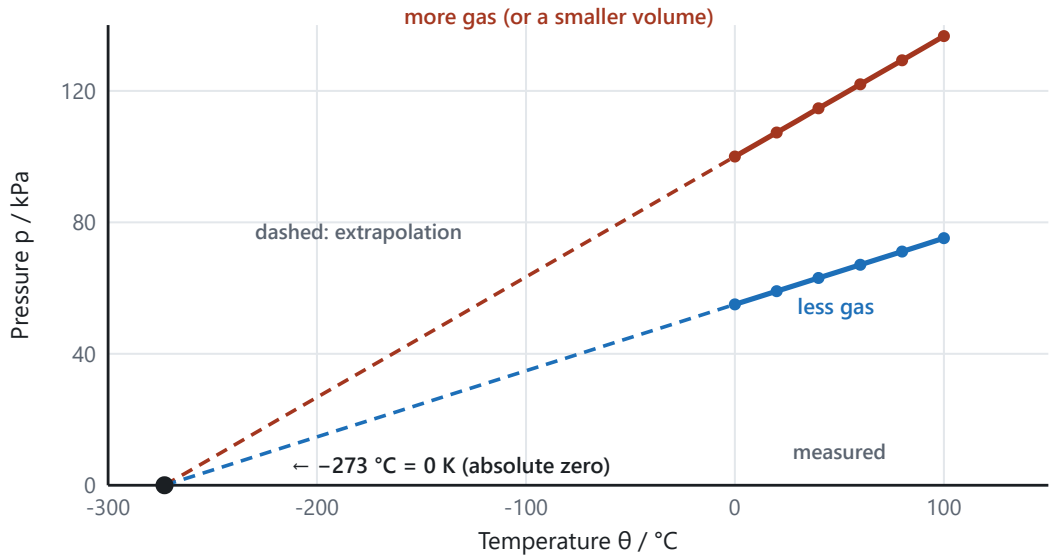
129 Absolute zero and the kelvin scale

- **Absolute zero** (0 K = −273 °C, more precisely −273.15 °C) is the **lowest possible temperature**: the molecules have their **minimum** kinetic energy (in the simple model, zero). Nothing can be colder, so there are no negative kelvin temperatures.
- Converting:

$$T / K = \theta / °C + 273$$

- The two scales have the same size of degree, so a **change** of 1 K equals a change of 1 °C.

129 Evidence from the gas laws



A fixed mass of gas at constant volume: p rises linearly with θ. Extrapolated back, every line reaches p = 0 at −273 °C, whatever the gas or its amount. That temperature is absolute zero.

- On the kelvin scale the same graph is a straight line **through the origin**: p ∝ T. That only works because the kelvin scale starts at absolute zero.

129 Mean KE is proportional to absolute temperature

- For an ideal gas the mean KE of a molecule is $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$ (spec point 132), so **mean KE ∝ T in kelvin**. Doubling T in **kelvin** doubles the mean KE; doubling the Celsius temperature does not.
- At the same temperature, **every** gas has the **same mean KE per molecule**. Lighter molecules therefore move faster.

129 Why heating a gas raises its pressure (the 6-mark chain)

1. Temperature rises, so the **mean kinetic energy** of the molecules rises.
2. So their mean (r.m.s.) **speed** rises, and their momentum rises.
3. Each collision with the wall gives a bigger **change of momentum**.
4. Molecules hit the walls **more often** (a higher rate of collision).
5. So the **rate of change of momentum** at the walls rises, which means a bigger **force** on the walls ($F = \Delta p/\Delta t$).
6. **Pressure** = force/area, so the pressure rises (volume fixed).

WORKED EXAMPLE: PRESSURE AND TEMPERATURE

A sealed tyre (fixed volume) is at 2.4×10^5 Pa at 12 °C. After a drive the air is at 35 °C. Find the new pressure.

$p \propto T$ at constant V and N, so $p_2 = p_1 \times T_2/T_1$

$p_2 = 2.4 \times 10^5 \times (35 + 273)/(12 + 273) = 2.4 \times 10^5 \times 308/285 =$ **2.6×10^5 Pa**

Using Celsius (35/12) would give a wildly wrong 7.0×10^5 Pa.

MARK-SCHEME LANGUAGE

- "average kinetic energy (of the molecules) increases", "(rms) speed increases" / "momentum increases", "greater change in momentum per collision", "more collisions per second with the walls", "rate of change of momentum increases so force increases", "p = F/A, so pressure increases".
- "Do not accept collisions between molecules": the collisions that matter are with the **walls** (WPH15 Jun 2021).

Often asked as: 6-mark "explain why the pressure falls when the football cools", "why the air tank's pressure changes", "why the balloon expands"; short 2–3 mark versions; MCQ: pressure quadruples, so T (in K) quadruples.

COMMON MISTAKE

Stopping at "the molecules move faster". You must go on to momentum change, collision rate, force and pressure. Saying the molecules collide with **each other** more. Not linking rate of change of momentum to force. Forgetting to use kelvin in a ratio.

BEYOND THE SPEC, BUT EXAMINED

- Comparing gases at the same temperature: same mean KE, so the lighter gas has the higher r.m.s. speed (WPH15 Jan 2021, Oct 2025).

SPEC 130

The ideal gas equation pV = NkT

WHAT THIS SPEC POINT ASKS

Use pV = NkT for an ideal gas.

130 The equation

$$pV = NkT$$

- p = pressure (Pa); V = volume (m^3); N = **number of molecules** (no unit); k = **Boltzmann constant** = 1.38 × 10^{-23} J K^{-1}; T = absolute temperature (K).
- k links the microscopic world (energy of one molecule) to temperature. It is tiny because one molecule carries very little energy.
- **Use this form, not pV = nRT.** The data booklet for this specification gives k, not R or N_A, and examiners report that the nRT route usually fails.

130 Converting units first

Quantity	Convert to
θ in °C	T = θ + 273 K
V in cm^3	× 10^{-6} m^3
V in dm^3 (litres)	× 10^{-3} m^3
p in kPa	× 10^3 Pa
mass of one molecule	A × 1.66 × 10^{-27} kg (A = relative molecular mass)

130 The gas laws that follow (fixed amount of gas, N constant)

- pV/T = Nk = constant, so

$$p_1V_1/T_1 = p_2V_2/T_2$$

- **Boyle's law**: constant T, p ∝ 1/V (spec point 131).
- **Charles's law**: constant p, V ∝ T.
- **Pressure law**: constant V, p ∝ T.
- At constant V and T, p ∝ N: let out half the molecules and the pressure halves.

130 What "ideal" means

An ideal gas obeys pV = NkT exactly. Kinetic theory models it with these assumptions: a very large number of molecules in **random motion**; the molecules' own volume is **negligible** compared with the container; **no forces** between molecules except during collisions; collisions are **elastic**; collision time is negligible compared with the time between collisions. Real gases behave nearly ideally at low pressure and high temperature.

WORKED EXAMPLE: NUMBER AND MASS OF MOLECULES

A 0.050 m^3 cylinder holds helium at 2.0×10^6 Pa and 17 °C. Find the number of molecules and the mass of helium (relative mass 4.0).
T = 17 + 273 = 290 K
N = pV/kT = $(2.0 \times 10^6 \times 0.050)/(1.38 \times 10^{-23} \times 290)$ = **2.5×10^{25}**
Mass of one atom = $4.0 \times 1.66 \times 10^{-27} = 6.64 \times 10^{-27}$ kg
Total mass = $2.5 \times 10^{25} \times 6.64 \times 10^{-27}$ = **0.17 kg**

WORKED EXAMPLE: A RISING BUBBLE

A 2.0 cm^3 air bubble is released 12 m below the surface of a lake where the water is at 8 °C. The surface water is at 18 °C. Atmospheric pressure = 1.01×10^5 Pa, ρ_{water} = 1000 kg m^{-3}. Find its volume at the surface.
$p_1 = p_{atm} + \rho gh = 1.01 \times 10^5 + 1000 \times 9.81 \times 12 = 2.19 \times 10^5$ Pa
$V_2 = V_1 \times (p_1/p_2) \times (T_2/T_1) = 2.0 \times (2.19 \times 10^5/1.01 \times 10^5) \times (291/281)$ = **4.5 cm^3**
(A ratio of volumes, so V can stay in cm^3 here; only the temperatures must be in kelvin.)

EXAM TIP

For "before and after" questions with a fixed amount of gas, the ratio form $p_1V_1/T_1 = p_2V_2/T_2$ is quickest. If gas leaks out, N changes: work out N before and after with N = pV/kT, then compare (e.g. percentage remaining).

MARK-SCHEME LANGUAGE

- "Use of pV = NkT [must see substitution of values of p, k and T]" (WPH15 Oct 2022).
- "Conversion of temperature to kelvin" is its own mark in nearly every question.
- "pV/T = constant" accepted for the ratio method.
- Absolute pressure at depth = atmospheric + hydrostatic (ρgh) (WPH15 Oct 2024).
- Final mark on a claim: "48 °C < 60 °C so the statement is inaccurate" (WPH15 Jan 2025).

Often asked as: p, V, N or T from the other three; number of molecules lost from a leak; mass of gas added; bubble rising from a lake bed; final temperature from a pressure ratio; gas density; balloon final temperature compared with a claim.

COMMON MISTAKE

Using pV = nRT. Confusing the Boltzmann constant k with the Coulomb constant k. Leaving T in °C. Leaving V in cm^3 when the equation needs m^3 (it only cancels in a pure ratio). Misreading what a given mass refers to (e.g. "2.00 g" being the total, not the added mass). Using $p \propto T$ and $V \propto T$ one after the other instead of the combined equation.

BEYOND THE SPEC, BUT EXAMINED

- **Gas density** $\rho = Nm/V = pm/kT$, where m is the mass of one molecule (WPH15 Oct 2022).
- **Gauge (excess) pressure** = absolute pressure − atmospheric pressure, e.g. for a football (WPH15 Oct 2020).
- Hydrostatic pressure $p = \rho gh$ from Unit 1 (WPH15 Oct 2024).
- **Balloons**: upthrust = weight of air displaced = $V\rho_{air}g$. A balloon rises when upthrust > weight and floats steadily when upthrust = weight. Upthrust is not caused by the helium molecules hitting the balloon (WPH15 Oct 2025).

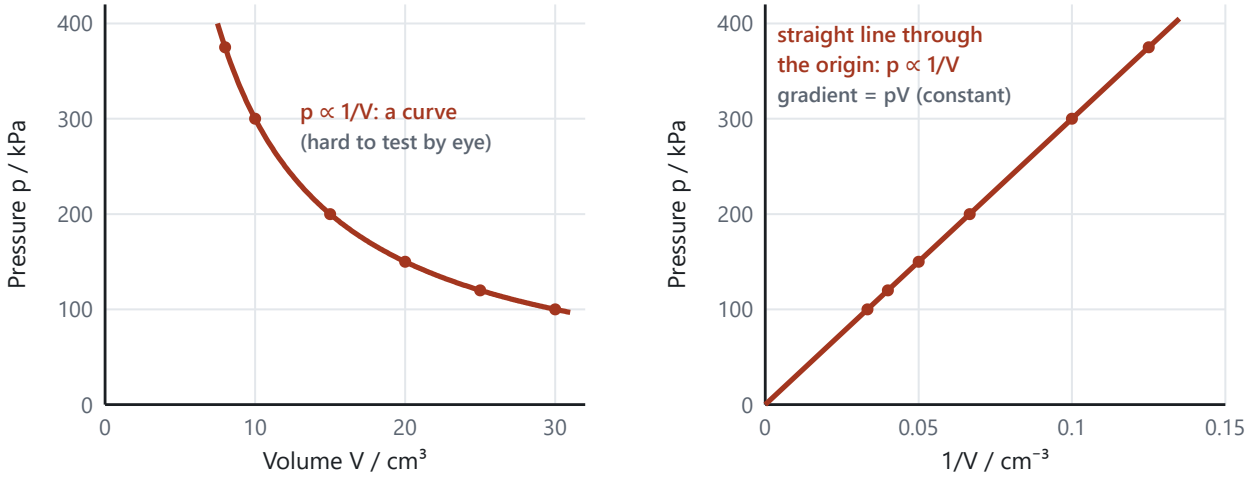
SPEC 131 Core practical 14: pressure and volume of a gas at constant temperature

WHAT THIS SPEC POINT ASKS

Investigate how the pressure of a fixed mass of gas depends on its volume at a fixed temperature (Boyle's law).

131 Expected result

$$pV = \text{constant} \quad \text{(fixed mass, fixed T)} \quad \text{so} \quad p \propto 1/V$$



Boyle's law data. A p–V graph is a curve that is hard to test by eye; plotting p against 1/V gives a straight line through the origin, which is an easy test.

CORE PRACTICAL 14: METHOD

Version A: Boyle's law apparatus. Air is trapped in a vertical glass tube above a column of oil. A foot pump pushes the oil up, compressing the air. A **Bourdon gauge** reads the pressure; the scale on the tube reads the volume (length × cross-sectional area, or volume directly).

1. Pump to the highest pressure, close the valve and **wait** for the air to return to room temperature (compressing it warms it).

2. Read p and V. Release a little pressure, wait again, and repeat for about 8 readings.
3. Read the oil level at eye level, at the bottom of the meniscus.

Version B: loaded syringe. Seal a syringe with some air in it and clamp it vertically, plunger up. Hang masses m from the plunger (or stand them on it). The pressure is $p = p_{atm} \pm mg/A$, where A is the plunger's cross-sectional area (from its diameter). Read V from the syringe scale; add the small volume trapped in the nozzle.

Variables: independent = volume (or the added mass); dependent = pressure; control = temperature, mass of gas (no leaks).
Processing: calculate pV for each pair, or plot p against 1/V. A straight line through the origin (or constant pV) confirms Boyle's law.
Safety: wear eye protection; don't exceed the apparatus's rated pressure; in version B keep feet clear of hanging masses.

WORKED EXAMPLE: TESTING A CLAIM THAT PV IS CONSTANT

Readings from a graph: (100 kPa, 30.0 cm^3), (150 kPa, 20.0 cm^3), (240 kPa, 12.6 cm^3).
pV: $100 \times 10^3 \times 30.0 \times 10^{-6} = 3.00$ Pa m^3; $150 \times 10^3 \times 20.0 \times 10^{-6} = 3.00$ Pa m^3; $240 \times 10^3 \times 12.6 \times 10^{-6} = 3.02$ Pa m^3
The three values agree to within 1%, so pV is constant and the claim is supported.

MARK-SCHEME LANGUAGE

- Take **at least two (better three) pairs** of p and V from the graph, calculate pV for each, and conclude that pV is constant "so the claim is valid" (WPH15 Oct 2023).
- "As volume increases pressure decreases" scores **nothing**: it doesn't show inverse proportion.

Often asked as: use a given p–V graph to test a claim; MCQ on how a change in method affects accuracy.

COMMON MISTAKE

Describing the trend instead of checking it with numbers. Compressing quickly and reading at once: the gas warms, so T isn't constant. Forgetting the dead volume in the syringe nozzle or gauge tubing, which makes every V too small and the p–1/V line miss the origin.

SPEC 132

Kinetic theory: mean kinetic energy and temperature

WHAT THIS SPEC POINT ASKS

Derive and use $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$, linking the mean kinetic energy of a gas molecule to absolute temperature.

132 Step 1: pressure from molecular collisions, $pV = \frac{1}{3}Nm\langle c^2\rangle$

Take N molecules, each of mass m, in a cube of side l (volume $V = l^3$), with the kinetic-theory assumptions listed under 130.

1. One molecule moves with velocity component c_x towards a wall. It bounces back elastically with $-c_x$. **Change of momentum** $= mc_x - (-mc_x) = \mathbf{2mc_x}$.
2. It travels 2l (across and back) before hitting the same wall again, so the **time between hits** is $2l/c_x$.
3. **Force** on the wall from this molecule = rate of change of momentum $= 2mc_x \div (2l/c_x) = mc_x^2/l$.
4. For all N molecules, use the **mean** of c_x^2: total force $F = Nm\langle c_x^2\rangle/l$.
5. **Pressure** $p = F/\text{area} = F/l^2 = Nm\langle c_x^2\rangle/l^3 = Nm\langle c_x^2\rangle/V$.

6. The motion is **random**, so no direction is special: $\langle c_x^2\rangle = \langle c_y^2\rangle = \langle c_z^2\rangle$. Since $c^2 = c_x^2 + c_y^2 + c_z^2$, $\langle c_x^2\rangle = \frac{1}{3}\langle c^2\rangle$.
7. So $p = \frac{1}{3}Nm\langle c^2\rangle/V$, or

$$pV = \frac{1}{3}Nm\langle c^2\rangle$$

- $\langle c^2\rangle$ is the **mean square speed** ($m^2\ s^{-2}$). Its square root is the **r.m.s. speed**, $c_{rms} = \sqrt{\langle c^2\rangle}$. Since Nm/V is the density ρ, this is also $p = \frac{1}{3}\rho\langle c^2\rangle$.

132 Step 2: combine with pV = NkT

- Both expressions equal pV: $\frac{1}{3}Nm\langle c^2\rangle = NkT$
- Cancel N and multiply by 3: $m\langle c^2\rangle = 3kT$
- Halve both sides:

$$\tfrac{1}{2}m\langle c^2\rangle = \tfrac{3}{2}kT$$

- The left side is the **mean kinetic energy of one molecule**. So mean KE ∝ T (kelvin), and it doesn't depend on which gas it is.
- Total KE of N molecules (the internal energy of an ideal gas) = $N \times \frac{3}{2}kT = \frac{3}{2}pV$.
- r.m.s. speed: $c_{rms} = \sqrt{(3kT/m)}$, so $c_{rms} \propto \sqrt{T}$ and, at a fixed T, $c_{rms} \propto 1/\sqrt{m}$.

WORKED EXAMPLE: MEAN KE AND R.M.S. SPEED

Nitrogen (N_2, relative molecular mass 28) at 20 °C.
T = 293 K
Mean KE = $\frac{3}{2}kT = 1.5 \times 1.38 \times 10^{-23} \times 293 =$ **6.07×10^{-21} J**
$m = 28 \times 1.66 \times 10^{-27} = 4.65 \times 10^{-26}$ kg
$\langle c^2\rangle = 3kT/m = (3 \times 1.38 \times 10^{-23} \times 293)/(4.65 \times 10^{-26}) = 2.61 \times 10^5\ m^2\ s^{-2}$
$c_{rms} = \sqrt{(2.61 \times 10^5)} =$ **$511\ m\ s^{-1}$**

WORKED EXAMPLE: COMPARING TWO GASES

Gas X molecules have an r.m.s. speed that is 60.5% of gas Y's at the same temperature. Find m_X/m_Y.
Same T, so same mean KE: $m_X\langle c_X^2\rangle = m_Y\langle c_Y^2\rangle$
$m_X/m_Y = \langle c_Y^2\rangle/\langle c_X^2\rangle = (1/0.605)^2 =$ **2.7**
Square the speed ratio: the 60.5% is of the r.m.s. speed, not of $\langle c^2\rangle$.

MARK-SCHEME LANGUAGE

- "Use of $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$"; for two gases at the same T, "mean KE is equal so $m_1\langle c_1^2\rangle = m_2\langle c_2^2\rangle$". "Do not award MP3 if a value for either mass has been assumed" (WPH15 Oct 2021).
- Derivation marks (when asked): change of momentum $2mc_x$; time between collisions $2l/c_x$; force = rate of change of momentum; $p = F/A$; $\langle c_x^2\rangle = \frac{1}{3}\langle c^2\rangle$ from random motion; then equate with pV = NkT.

Often asked as: mean KE per molecule; ratio of masses or speeds from r.m.s. speeds (MCQ ratio √3); internal energy $N \times \frac{3}{2}kT$; derive $U = \frac{3}{2}pV$. In WPH15 the derivation has so far been "combine pV = NkT with $pV = \frac{1}{3}Nm\langle c^2\rangle$", but learn the full collision derivation too.

COMMON MISTAKE

Using the r.m.s. speed ratio as if it were the mean-square ratio (not squaring). Multiplying by N when the question asks for one molecule. Leaving T in °C. Thinking a heavier gas has more KE per molecule at the same temperature: it has the same KE and a lower speed.

BEYOND THE SPEC, BUT EXAMINED

- Molecular speeds in a gas have a **range** (a distribution), so even when c_{rms} is below the escape speed, some molecules are fast enough to leave a planet's atmosphere (WPH15 Jan 2023, Jan 2026). Light gases (hydrogen, helium) escape most easily.

Topic 5.4 – Nuclear Decay

SPEC 133 Binding energy and mass deficit

WHAT THIS SPEC POINT ASKS

Understand nuclear binding energy, and use $\Delta E = c^2\Delta m$ to calculate mass deficits and the energy released in nuclear processes.

133 Mass deficit

- A nucleus has **less mass** than the protons and neutrons (nucleons) that make it, measured separately. The difference is the **mass deficit** (mass defect) Δm:

$$\Delta m = Zm_p + (A - Z)m_n - m_{nucleus}$$

- Z = proton number; A = nucleon number; m_p, m_n = proton and neutron masses.

133 Binding energy

- The **binding energy** of a nucleus is the energy needed to **separate it completely into its individual nucleons**. Equivalently, it is the energy released when the nucleus forms from separate nucleons.
- Mass and energy are equivalent (Einstein):

$$\Delta E = c^2\Delta m$$

- ΔE = energy (J); c = 3.00×10^8 m s^{-1}; Δm = mass (kg).
- Binding energy is **not** energy stored inside the nucleus. A nucleus with a large binding energy is **more** tightly bound: you'd have to put more energy in to pull it apart.
- **Binding energy per nucleon** = total binding energy ÷ A. The higher it is, the more stable the nucleus (spec point 135).

133 Energy released in a decay or reaction

- In any spontaneous decay, fission or fusion, the **products have less total mass** than the starting particles. The missing mass appears as energy (mostly KE of the products, plus gamma photons):

$$\text{energy released} = c^2 \times (\text{mass before} - \text{mass after})$$

133 The calculation chain (each step is a mark)

1. Find Δm in **u** (total mass before − total mass after; or nucleons − nucleus).
2. Convert to kg: × 1.66×10^{-27} kg u^{-1}.
3. $\Delta E = c^2\Delta m$ in J.
4. Convert to MeV: ÷ 1.60×10^{-13} J MeV^{-1} (or to eV: ÷ 1.60×10^{-19}).
5. For binding energy per nucleon, divide by A.

Keep at least 5–6 significant figures in the masses until after the subtraction: the difference is tiny.

WORKED EXAMPLE: BINDING ENERGY PER NUCLEON OF IRON-56

Fe-56 has 26 protons and 30 neutrons. Mass of the nucleus = 55.92067 u; m_p = 1.00728 u; m_n = 1.00867 u.

$\Delta m = 26 \times 1.00728 + 30 \times 1.00867 - 55.92067 = 26.18928 + 30.26010 - 55.92067 = 0.52871$ u

$\Delta m = 0.52871 \times 1.66 \times 10^{-27} = 8.777 \times 10^{-28}$ kg

$\Delta E = c^2\Delta m = (3.00 \times 10^8)^2 \times 8.777 \times 10^{-28} = 7.90 \times 10^{-11}$ J

In MeV: $7.90 \times 10^{-11} / 1.60 \times 10^{-13} = 494$ MeV

Per nucleon: 494/56 = **8.8 MeV**

WORKED EXAMPLE: ENERGY RELEASED IN ALPHA DECAY

$^{226}Ra \rightarrow {}^{222}Rn + {}^{4}He$. Atomic masses: Ra-226 = 226.025410 u, Rn-222 = 222.017578 u, He-4 = 4.002603 u.

$\Delta m = 226.025410 - (222.017578 + 4.002603) = 0.005229$ u

$= 0.005229 \times 1.66 \times 10^{-27} = 8.680 \times 10^{-30}$ kg

$\Delta E = (3.00 \times 10^8)^2 \times 8.680 \times 10^{-30} = 7.81 \times 10^{-13}$ J = **4.88 MeV**

Atomic masses work here because the electrons balance on both sides (88 = 86 + 2).

EXAM TIP

Write down the mass difference in u **before** converting. Examiners give the subtraction its own mark, and "a bald answer scores zero": you need at least two of the steps written (WPH15 Oct 2021). Don't round in the middle: rounding Δm early turned 7.88 MeV into 7.85 MeV and lost the mark (WPH15 Jan 2026).

MARK-SCHEME LANGUAGE

- "Use of $\Delta E = c^2\Delta m$", "conversion of u to kg" ($\times 1.66 \times 10^{-27}$), "conversion of J to MeV".
- "Do not accept use of 1.67×10^{-27}" for u (WPH15 Jan 2026); use the booklet value 1.66×10^{-27} kg.
- Binding energy per nucleon: give at least 2 s.f., worked from the masses given in the question without early rounding; an answer that drifts from the data value loses the final mark (WPH15 Oct 2025).
- Binding energy: "energy needed to separate a nucleus into its (individual) nucleons".

Often asked as: energy released in an alpha or beta decay from atomic masses; binding energy (per nucleon) from nucleon masses; mass converted to energy per second in the Sun; MCQ on what binding energy means.

COMMON MISTAKE

Applying $E = mc^2$ to the whole mass of a nucleus instead of the mass **difference**. Subtracting in the wrong order, or leaving out one of the three masses. Multiplying nuclide masses by the number of nucleons. Forgetting the 10^{-27}. Using the proton mass instead of u to convert. Dividing when you should multiply (J → MeV is $\div 1.60 \times 10^{-13}$). Forgetting to divide by A for "per nucleon". Giving a decay energy when the binding energy of a nucleus was asked for.

BEYOND THE SPEC, BUT EXAMINED

- **Beta decay with atomic masses**: in β^- decay the daughter atom has one more electron than the parent's nucleus needs, so atomic masses already include the emitted electron's mass. Check what the question gives: one paper subtracted the electron mass (0.00055 u) explicitly (WPH15 Jan 2021).
- Masses in unfamiliar units such as **GeV/c²**: multiply by $10^9 \times 1.60 \times 10^{-19}$ to get J directly (WPH15 Jun 2021).
- The energy released is shared between the products by **momentum conservation** (see 139).

SPEC 134 The atomic mass unit

WHAT THIS SPEC POINT ASKS

Express small masses in atomic mass units (u) and convert between u and kg.

134 Definition and conversions

- The **atomic mass unit** u is **one twelfth of the mass of a carbon-12 atom**. It is roughly the mass of one nucleon.

$$1\ \text{u} = 1.66 \times 10^{-27}\ \text{kg} \quad \text{(data booklet)}$$

- The mass of a nucleus in u is close to (slightly less than) its nucleon number A. So a quick estimate of the mass of one nucleus is $A \times 1.66 \times 10^{-27}$ kg.
- Energy equivalent of 1 u: $E = c^2\Delta m = (3.00 \times 10^8)^2 \times 1.66 \times 10^{-27} = 1.494 \times 10^{-10}$ J = **934 MeV** with booklet values (the precise value is 931.5 MeV). You may use "1 u ≈ 934 MeV" as a shortcut if you show where it comes from.
- 1 eV = 1.60×10^{-19} J; 1 MeV = 1.60×10^{-13} J.

WORKED EXAMPLE: MASS OF NUCLEI FROM A COUNT

A sample contains 3.0×10^{21} uranium-238 nuclei. Find the mass of uranium.
Mass of one nucleus ≈ $238 \times 1.66 \times 10^{-27} = 3.95 \times 10^{-25}$ kg
Total = $3.0 \times 10^{21} \times 3.95 \times 10^{-25}$ = **1.2×10^{-3} kg** (1.2 g)

MARK-SCHEME LANGUAGE

- "$\times 1.66 \times 10^{-27}$ kg u^{-1}" is the conversion mark in every decay-energy question.
- "Mass of one nucleus = A × u" for an approximate mass; "use of 1 u = 934 MeV (ecf)" accepted (WPH15 Jan 2022).

Often asked as: the conversion step inside 133; mass of a sample from the number of nuclei (often from N = A/λ in 142).

COMMON MISTAKE

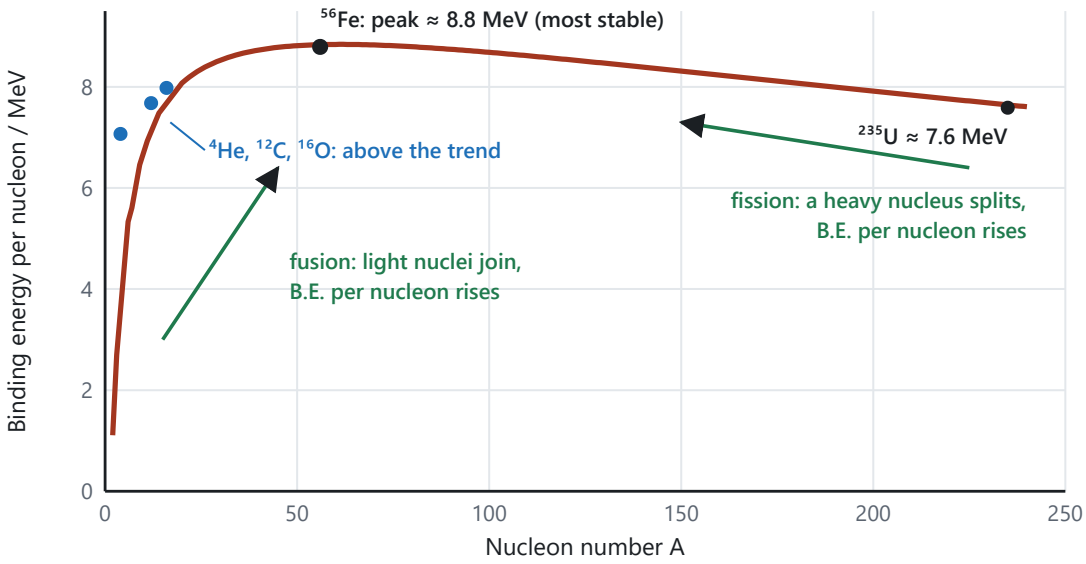
Using 1.67×10^{-27} kg (the proton mass) instead of 1.66×10^{-27} kg. Using $A \times m_p$ as the mass of a nucleus.

SPEC 135 Fission, fusion and the binding energy per nucleon curve

WHAT THIS SPEC POINT ASKS

Explain nuclear fission and fusion using the graph of binding energy per nucleon against nucleon number.

135 The curve



Binding energy per nucleon against nucleon number A. It rises steeply for light nuclei, peaks at about 8.8 MeV near iron-56, then falls slowly to about 7.6 MeV for uranium. He-4, C-12 and O-16 lie above the general trend.

- **Iron-56** (and its neighbours) has the **highest binding energy per nucleon**: it is the most stable nucleus.
- Any change that **moves nuclei up the curve** (towards iron) **releases energy**, because the products are more tightly bound and so have **less total mass**.

135 Fusion

- **Fusion**: two **light** nuclei join to make a heavier nucleus. For light nuclei (left of the peak) the binding energy per nucleon **increases**, so energy is released.
- The curve is steepest here, so fusion releases more energy **per kilogram of fuel** than fission.

135 Fission

- **Fission**: a **heavy** nucleus (e.g. U-235, after absorbing a slow neutron) **splits** into two medium-sized nuclei plus 2 or 3 neutrons. For nuclei right of the peak, the products have a **higher binding energy per nucleon**, so energy is released.
- The released neutrons can cause further fissions: a **chain reaction**.
- Fission of a nucleus **lighter** than iron (e.g. He-4) would **absorb** energy, because the products would have a lower binding energy per nucleon.

135 Energy released from graph readings

$$\text{energy released} = (\text{total binding energy of products}) - (\text{total binding energy of reactants})$$

with total binding energy = A × (binding energy per nucleon) for each nucleus.

WORKED EXAMPLE: ENERGY FROM FUSION AND FISSION USING THE CURVE

Fusion $^{2}H + {}^{3}H \rightarrow {}^{4}He + n$. Binding energies per nucleon: ^{2}H 1.11 MeV, ^{3}H 2.83 MeV, ^{4}He 7.07 MeV (a lone neutron has none).
Reactants: $2 \times 1.11 + 3 \times 2.83 = 10.71$ MeV
Products: $4 \times 7.07 = 28.28$ MeV
Released = $28.28 - 10.71 =$ **17.6 MeV**
Fission $^{235}U + n \rightarrow {}^{141}Ba + {}^{92}Kr + 3n$. Readings: U-235 7.6 MeV, Ba-141 8.3 MeV, Kr-92 8.5 MeV.
Released = $(141 \times 8.3 + 92 \times 8.5) - 235 \times 7.6 = 1952 - 1786 =$ **about 170 MeV**
Per nucleon of fuel: fusion $17.6/5 \approx 3.5$ MeV; fission $170/236 \approx 0.7$ MeV. Fusion gives more energy per kilogram.

EXAM TIP

To sketch the curve for marks: start near the origin and rise **steeply**, peak at **Fe-56** (label it, about 8–9 MeV), then fall **slowly**. Label the y-axis "binding energy **per nucleon** / MeV". Read a total binding energy from the graph as A × (the y-value at that A): the superscript in the symbol is the A you need.

MARK-SCHEME LANGUAGE

- Fission: "a large nucleus splits (into two (smaller) nuclei and neutrons)"; "binding energy (per nucleon) increases so energy is released", or "decrease in (total) mass so energy is released" (WPH15 Jan 2024).
- Fusion: "light nuclei combine"; "binding energy per nucleon increases".
- Graph sketch: "steeply rising curve near to origin; slowly decreasing curve after peak; iron-56 marked at peak" (WPH15 Jun 2021).
- Fe-56 has the greatest binding energy **per nucleon**, not the greatest total binding energy (WPH15 Jan 2023).

Often asked as: define fission; sketch the curve and mark Fe-56; balance a fission equation; explain how fission or fusion releases energy; energy released from graph readings; compare Fe and U binding energies.

COMMON MISTAKE

"Binding energy decreases in fission, so energy is released" is wrong: it **increases**. Using the word "fission" without saying what happens. Confusing total and per-nucleon binding energy. Only counting one fission product's binding energy.

BEYOND THE SPEC, BUT EXAMINED

- Balancing a fission equation including the neutrons: $^{236}U \rightarrow {}^{93}Sr + {}^{141}Xe + 2n$ (WPH15 Jun 2021).
- A trap: fission of He-4 would **decrease** binding energy per nucleon, so it would not release energy (WPH15 Oct 2025).
- Estimating U-235's binding energy per nucleon from the graph by proportion (about 7.4–7.6 MeV) (WPH15 Oct 2025).

SPEC 136

Conditions for nuclear fusion

WHAT THIS SPEC POINT ASKS

Understand why fusion needs very high temperatures and very high densities to start and to keep going.

136 Why it is hard

- Nuclei are **positively charged**, so they **repel** each other electrostatically. The repulsion grows as they get closer.
- The **strong nuclear force** that can bind them together only acts over a **very short range** (about 10^{-15} m, the size of a nucleus).
- To fuse, two nuclei must get **close enough for the strong force to act**. They must approach with a very large kinetic energy to overcome the repulsion.

136 The two conditions

1. **Very high temperature** (about 10^7 K in the Sun's core; around 10^8 K in Earth reactors): mean KE $\propto$ T, so the nuclei move fast enough to overcome the electrostatic repulsion. At these temperatures matter is a **plasma**: electrons are stripped from the nuclei.
2. **Very high density** (a large number of nuclei per unit volume): collisions happen **often enough** to keep the fusion rate high, so enough energy is released to **maintain** the high temperature and sustain fusion.

136 Where these happen

- In a **star**, gravity squeezes the core to a huge density, and the gravitational collapse raises the temperature.
- On **Earth**, the plasma must be **confined** by strong **magnetic fields**: it can't touch a solid wall, which would cool it and stop fusion (and would be damaged). Keeping the plasma hot, dense and confined for long enough is why fusion power is so difficult.
- In the Sun, hydrogen fuses to helium (net: 4 protons → 1 helium-4 nucleus + 2 positrons + 2 neutrinos + energy).

WORKED EXAMPLE: THE SUN'S MASS LOSS

The Sun's luminosity is 3.85×10^{26} W. How much mass does it convert to energy each second?
$\Delta m/\Delta t = L/c^2 = 3.85 \times 10^{26} / (3.00 \times 10^8)^2 =$ **4.3×10^9 kg s^{-1}**

MARK-SCHEME LANGUAGE

The indicative content is the same in every paper (WPH15 Oct 2021, Jan 2024, Oct 2024, Jan 2026, Jun 2026):

- (very) **high temperature**;
- so the nuclei/protons have enough **kinetic energy** ("kinetic energy, not just energy");
- to **overcome the electrostatic repulsion** between the positively charged nuclei;
- so the nuclei get **close enough to fuse** (within range of the strong force);
- (very) **high density**;
- so there is a **high collision rate**, to sustain fusion / maintain the temperature.
  "Not 'particles', not 'molecules', not 'atoms'": say **nuclei** or **protons**. Accept "collide more often"; reject "number of collisions" (WPH15 Jun 2026).

Often asked as: a 4-mark short answer or a 6-mark linkage question on fusion in a star's core or a reactor; why Earth fusion needs magnetic confinement; mass lost per second by the Sun.

COMMON MISTAKE

Writing "particles" or "atoms" instead of nuclei. Saying high density means "more collisions" without "per second" (rate). Answering about the balance of forces inside a star (that's a different question). Saying "energy" when the mark needs "kinetic energy".

BEYOND THE SPEC, BUT EXAMINED

- Plasma containment on Earth with strong magnetic fields (WPH15 Oct 2020).
- Mass converted per second from L/c^2, and the fraction of the Sun's mass used up over a time (WPH15 Oct 2020, Jun 2024).

SPEC 137

Background radiation

WHAT THIS SPEC POINT ASKS

Know that background radiation is always present and correct measured count rates for it.

137 Sources

- **Natural**: **radon** gas seeping from rocks and soil (the largest share in many places; an alpha emitter, hazardous when breathed in); rocks and buildings (granite); **cosmic rays** from space; food and drink (potassium-40, carbon-14), so our own bodies.
- **Artificial**: medical X-rays and tracers (the biggest artificial share); fallout from past weapons tests and accidents; nuclear industry (very small).
- It varies with location (rock type, altitude).

137 Correcting for it

1. With **no source present**, measure the count over a **long time** (several minutes) and work out the background count rate.
2. **Subtract** it from **every** reading taken with the source:

$$\text{corrected count rate} = \text{measured count rate} - \text{background count rate}$$

3. Do the subtraction **before** any further processing (logs, ratios, half-lives).

- A decay curve that levels off **above zero**, or a count that never falls below a steady value, is showing the background.
- Radioactive counts fluctuate randomly, so longer counting times give more reliable averages.

WORKED EXAMPLE: CORRECTED COUNT RATE

Background: 270 counts in 10 minutes. With the source: 1950 counts in 5 minutes. Find the corrected count rate.
Background rate = 270/10 = 27 counts per minute
Measured rate = 1950/5 = 390 counts per minute
Corrected = 390 − 27 = **363 counts per minute**

MARK-SCHEME LANGUAGE

- "Measure the count (rate) with no source present (over a long time)"; "subtract (the background count rate) from each reading".
- Without the correction "the count rate is overestimated" / "background count introduces a **systematic** error". Just saying "it gives a more accurate value" scored nothing (WPH15 Jun 2024).

Often asked as: MCQs on how background affects a measurement (WPH15 Jun 2021, Jan 2026); deduce whether an activity has fallen to background level; describe how to find the background and why it causes a systematic error.

COMMON MISTAKE

Subtracting the background from the **initial** activity in a calculation where the question's numbers are already corrected. Muddling initial and final activities when finding the time to fall to background.

SPEC 138 Alpha, beta and gamma radiation

WHAT THIS SPEC POINT ASKS

Relate the nature of alpha, beta and gamma radiation to their ionising ability, penetration and range in different materials.

138 Properties

	Alpha (α)	Beta-minus (β^-)	Gamma (γ)
Nature	helium nucleus: 2 protons + 2 neutrons	fast **electron** from the nucleus	high-energy **electromagnetic** wave (photon)
Charge	**+2e**	**−e**	0
Mass	about 4 u	about 1/1840 u	0
Ionising ability	**strong**	moderate	weak
Range in air	a few cm	tens of cm to about 1 m	very long (intensity falls with distance)
Stopped by	a sheet of paper, or skin	a few mm of aluminium	reduced (never fully stopped) by several cm of lead or metres of concrete
Deflection in fields	small (heavy), towards the negative plate	large (light), opposite way to α	not deflected

(β^+ is a **positron**, charge **+e**; its properties are otherwise like β^-, but it annihilates with an electron almost at once.)

138 Ionising ability links to range

- **Ionisation**: the radiation knocks electrons out of atoms it passes, losing some energy each time.
- **Alpha** is massive and doubly charged, so it interacts strongly and ionises densely. It loses its energy fast, so it has a **short range**: **the more strongly ionising, the shorter the range**.
- **Gamma** is uncharged and interacts rarely, so it ionises weakly and penetrates far. Its intensity falls **exponentially** with absorber thickness (spec point 140), so there is no definite range.

138 Safety

- Outside the body, alpha is the **least** dangerous (skin stops it). **Inside** the body (inhaled or swallowed) it is the **most** dangerous, because all its ionisation is done in a small volume of living tissue.
- Handle sources with **tongs** (distance), for as **short a time** as possible ("the ionising effect is cumulative"), with **shielding**; store in lead-lined boxes.

MARK-SCHEME LANGUAGE

- Cloud chamber: "the thickness of the track is related to the ionising ability: alpha is strongly ionising and beta only moderately ionising (so alpha tracks are thick, beta tracks thin)"; "the shape of the track is related to the mass: alpha particles are massive (so straight tracks) and beta are not (so twisted tracks)" (WPH15 Jun 2023).
- Alpha: "can only travel a few cm in air", "only a danger if taken into the body" (WPH15 Jun 2026).
- Safety: "use tongs", "minimise time of exposure as the ionising effect is cumulative" (WPH15 Oct 2020).

Often asked as: comment on each radiation in turn in a given context; cloud-chamber tracks; safety with a source; shielding thickness from a graph; tracer choice.

COMMON MISTAKE

GCSE-style lists of materials (lead, concrete) that ignore the context in the question. Not commenting on each radiation separately. Confusing "ionisation energy" with ionising ability. Linking a twisting track to speed rather than to the particle's small **mass**. Giving beta a charge of +1e: β^- is −e (only the positron, β^+, is +e).

BEYOND THE SPEC, BUT EXAMINED

- **PET scans**: a β^+ tracer emits positrons that **annihilate** with electrons, producing two gamma photons that pass out of the body to the detectors. The tracer's half-life must be long enough to do the scan but short enough to limit the dose (WPH15 Oct 2022).
- An alpha particle ionising air excites molecules, which emit **UV photons** as they de-excite; the weakly ionising UV travels much further, so it can be detected at a safe distance (WPH15 Jun 2021).
- The fraction of beta absorbed rises with paper thickness: quote the relationship, not vague "absorbs more" (WPH15 Jan 2025).

SPEC 139

Nuclear equations

WHAT THIS SPEC POINT ASKS

Write and interpret nuclear equations for decays and reactions, given the particle symbols.

139 Rules

- Write each nucleus as ${}^{A}_{Z}X$ (A = nucleon number on top, Z = proton number below).
- **Nucleon number (top) is conserved** and **charge/proton number (bottom) is conserved**: the totals are the same on both sides.

139 The particle symbols

Particle	Symbol	Top (A)	Bottom (Z)
alpha	$^{4}_{2}\alpha$ or $^{4}_{2}\mathrm{He}$	4	2
beta-minus (electron)	$^{0}_{-1}\beta$ or $^{0}_{-1}\mathrm{e}$	0	−1
beta-plus (positron)	$^{0}_{+1}\beta$ or $^{0}_{+1}\mathrm{e}$	0	+1
neutron	$^{1}_{0}\mathrm{n}$	1	0
proton	$^{1}_{1}\mathrm{p}$	1	1
gamma	$^{0}_{0}\gamma$	0	0
electron antineutrino / neutrino	$\bar{\nu}_e$ / ν_e	0	0

139 The decays

- **Alpha**: A falls by 4, Z falls by 2.
  $^{226}_{88}\mathrm{Ra} \rightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\alpha$
- **Beta-minus**: a neutron becomes a proton ($n \rightarrow p + e^{-} + \bar{\nu}_e$). A unchanged, Z **rises** by 1. An **antineutrino** is emitted too.
  $^{90}_{38}\mathrm{Sr} \rightarrow {}^{90}_{39}\mathrm{Y} + {}^{0}_{-1}\beta + \bar{\nu}_e$
- **Beta-plus**: a proton becomes a neutron ($p \rightarrow n + e^{+} + \nu_e$). A unchanged, Z **falls** by 1. A **neutrino** is emitted.
  $^{18}_{9}\mathrm{F} \rightarrow {}^{18}_{8}\mathrm{O} + {}^{0}_{+1}\beta + \nu_e$
- **Gamma**: the nucleus drops from an excited state to a lower energy state. A and Z don't change. It often follows alpha or beta decay.
- **Fission** and **fusion** obey the same rules: $^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3\,{}^{1}_{0}\mathrm{n}$ (top: 236 = 141 + 92 + 3; bottom: 92 = 56 + 36).

139 How the energy is shared

- **Alpha decay** (two products): momentum is conserved, so the alpha and the daughter nucleus move off with **equal and opposite momentum**. Since $E_k = p^2/2m$, the much lighter alpha gets almost all the KE, and every alpha from that decay has the **same energy**.
- **Beta decay** (three products: daughter, electron, antineutrino): the energy can be shared in many ways, so beta particles have a **continuous range of energies** up to a maximum. This was the evidence for the neutrino.

WORKED EXAMPLE: KE OF THE ALPHA PARTICLE

In the Ra-226 decay 4.88 MeV is released. The alpha (4 u) and radon (222 u) share it. Find the alpha's KE.
Equal momentum p, and $E_k = p^2/2m$, so $E_k \propto 1/m$.
$E_\alpha = 4.88 \times 222/(222 + 4) =$ **4.79 MeV** (the radon gets only 0.09 MeV)

MARK-SCHEME LANGUAGE

- Top line (nucleon numbers) and bottom line (proton numbers) are separate marks.
- Beta-minus: electron as $^{0}_{-1}e$ "plus an antineutrino"; the daughter has **Z + 1**.
- Range of beta energies: "momentum/KE is **shared between three** particles"; "the energy split between the beta particle and the (anti)neutrino is random" (WPH15 Jan 2021).
- Alpha: "momentum is conserved, so the velocity of the heavy nucleus is much less, so its KE is much smaller than that of the alpha" (WPH15 Oct 2021).

Often asked as: complete a β^- equation with the antineutrino (Sr-90, tritium, K-40); balance alpha, fission or fusion equations; why beta energies vary; why the recoiling nucleus gets almost no KE.

COMMON MISTAKE

Writing the beta particle as A = −1, Z = 0 (upside down). Subtracting 1 from Z in β^- decay, or combining −1 and 38 to give 37 for Y-90 (it's 39). Forgetting the antineutrino. Leaving out the neutrons in a fission equation.

SPEC 140

Core practical 15: absorption of gamma radiation by lead

WHAT THIS SPEC POINT ASKS

Investigate how the count rate of gamma radiation depends on the thickness of lead absorber.

140 The relationship

- Gamma intensity (and corrected count rate R) falls **exponentially** with absorber thickness x:

$$R = R_0 e^{-\mu x}$$

- R_0 = corrected count rate with no absorber; μ = **attenuation (absorption) coefficient** (unit: cm^{-1} or m^{-1}, the inverse of x's unit).
- Take natural logs to get a straight line:

$$\ln R = \ln R_0 - \mu x$$

- Plot ln R (y) against x: **gradient = −μ**, y-intercept = $\ln R_0$.
- **Half-value thickness** (thickness that halves the count rate): $x_{½} = \ln 2/\mu$.

CORE PRACTICAL 15: METHOD

1. With no source nearby, record the **background** count over at least 5 minutes; find the background count rate.
2. Clamp the gamma source and the Geiger–Müller tube a fixed distance apart (e.g. 10 cm). Record the count over a fixed time (e.g. 1 minute) with no lead.
3. Measure each lead sheet's thickness with a **micrometer** (or vernier calipers) at several points and average.
4. Put lead sheets between source and tube, building up the thickness. For each thickness take **three** 1-minute counts and average.
5. Subtract the background from every count rate, then plot ln(corrected count rate) against thickness.

Variables: independent = lead thickness; dependent = count rate; control = source–detector distance, same source and detector, counting time, position (same background).
Safety: lift the source with **long tongs**, point it away from people, keep it at arm's length and return it to

its lead-lined box straight away; minimise time near it; wash hands after handling lead.
Improvements: count for longer to reduce the effect of random fluctuations; keep the distance fixed with a clamp and ruler.

WORKED EXAMPLE: ATTENUATION COEFFICIENT AND HALF-VALUE THICKNESS

Corrected count rates (counts per minute): x = 0 cm, 480; 0.5 cm, 374; 1.0 cm, 291; 1.5 cm, 227; 2.0 cm, 177; 2.5 cm, 138.
ln R: 6.17, 5.92, 5.67, 5.42, 5.17, 4.92 (a straight line, so the decay is exponential).
Gradient (large triangle, ends of the line) = (4.92 − 6.17)/(2.5 − 0) = −0.50 cm^{-1}, so **$\mu = 0.50\ cm^{-1}$**
$x_{½} = \ln 2/\mu = 0.693/0.50 =$ **1.4 cm**
Check: R at 1.4 cm ≈ 480 × $e^{-0.70}$ ≈ 240, half of 480.

MARK-SCHEME LANGUAGE

- "Log expansion of $R = R_0e^{-\mu x}$", "μ identified as (−) gradient", "gradient calculated", "large triangle used", "conclusion consistent with the half-value thickness" (WPH15 Jan 2021).
- Give μ **with its unit** (cm^{-1} or m^{-1}): many correct values lost the unit mark (WPH15 Jun 2024).
- From a curve, μ = 1/(the intercept the tangent at x = 0 makes on the x-axis) (WPH15 Jun 2024).

Often asked as: process log data to find μ and compare a half-value thickness with a claim; read μ from a graph; percentage transmission and shield thickness; background correction MCQs; safety.

COMMON MISTAKE

Taking a gradient on a log-scale plot of the raw data instead of on a ln R against x graph. Forgetting to subtract background before taking logs. Spending time on distractor data (such as the gamma energy) in the stem. Power-of-ten errors when converting cm and m.

BEYOND THE SPEC, BUT EXAMINED

- The equation $R = R_0e^{-\mu x}$ and the term "attenuation coefficient" are not in the spec, but appear in the question stem and must be used (WPH15 Jan 2021).
- Reading values from a log scale (the y-axis printed with 10, 100, 1000…) (WPH15 Jan 2022, Jan 2025).

SPEC 141 The random and spontaneous nature of radioactive decay

WHAT THIS SPEC POINT ASKS

Understand that radioactive decay is spontaneous and random.

141 The two words

- **Random**: you **cannot predict which** nucleus will decay next, or **when** any particular nucleus will decay. Each nucleus has the same, constant **probability** of decaying per unit time (the decay constant λ), regardless of how long it has existed.
- **Spontaneous**: the decay is **not affected by external factors**: not temperature, pressure, chemical combination, or the presence of other nuclei. Nothing triggers it.

141 Consequences

- Because every nucleus has the same probability λ per second, the number decaying per second is proportional to the number present: **dN/dt = −λN** (spec point 142). That is why decay is **exponential** and the half-life is constant.
- With huge numbers of nuclei, the **average** behaviour is very predictable, even though each decay is random.
- **Evidence**: counts recorded over equal time intervals **fluctuate** about a mean; the clicks of a Geiger counter are irregular. The spread is about √N for N counts, so longer counts give smaller percentage uncertainty.

MARK-SCHEME LANGUAGE

- Random: "it is not possible to predict **which** nucleus will decay next, or **when**".
- Spontaneous: "not affected by external factors such as temperature or pressure" (or by the presence of other nuclei).
- The false statement in a 2026 MCQ was "it is possible to predict which nucleus will decay next" (WPH15 Jun 2026).

Often asked as: MCQs testing the two definitions (only about a third answered the WPH15 Jan 2024 one correctly); used as the justification for dN/dt = −λN and for fluctuating counts.

COMMON MISTAKE

Swapping the meanings: "random" is about prediction; "spontaneous" is about outside influences. Saying the half-life of a single nucleus is predictable (only the behaviour of a large number is).

SPEC 142 Half-life and the decay equations

WHAT THIS SPEC POINT ASKS

Find half-lives graphically and use A = λN, dN/dt = −λN, λ = ln2/t½, $N = N_0e^{-\lambda t}$ and $A = A_0e^{-\lambda t}$; derive and use the corresponding log equations.

142 Activity and the decay constant

- **Activity** A = number of decays per second, unit **becquerel** (Bq = s^{-1}).
- **Decay constant** λ = probability that a given nucleus decays per unit time (s^{-1}, or h^{-1}, y^{-1}...).

$$A = \lambda N \qquad dN/dt = -\lambda N$$

- The minus sign shows N **decreases**. A is the size of dN/dt.
- For A in Bq, λ must be in **s^{-1}**.

142 Exponential decay

$$N = N_0e^{-\lambda t} \qquad A = A_0e^{-\lambda t}$$

- N_0, A_0 = values at t = 0. λ and t must be in **matching** time units (λ in day^{-1} with t in days is fine).
- The mass of the parent and the corrected count rate fall in the same way (both ∝ N).

142 Half-life

- **Half-life** t½: the **time taken for the number of undecayed nuclei (or the activity) to halve**. It is constant for a given isotope.

- Derivation: put $N = N_0/2$ at $t = t_{½}$:
  $N_0/2 = N_0e^{-\lambda t_{½}} \rightarrow ½ = e^{-\lambda t_{½}} \rightarrow e^{\lambda t_{½}} = 2 \rightarrow \lambda t_{½} = \ln 2$, so

$$t_{½} = \ln 2/\lambda \approx 0.693/\lambda$$

- After n half-lives the fraction left is $(½)^n$. This works for whole numbers of half-lives; for anything else (e.g. "10% remaining") use logs.

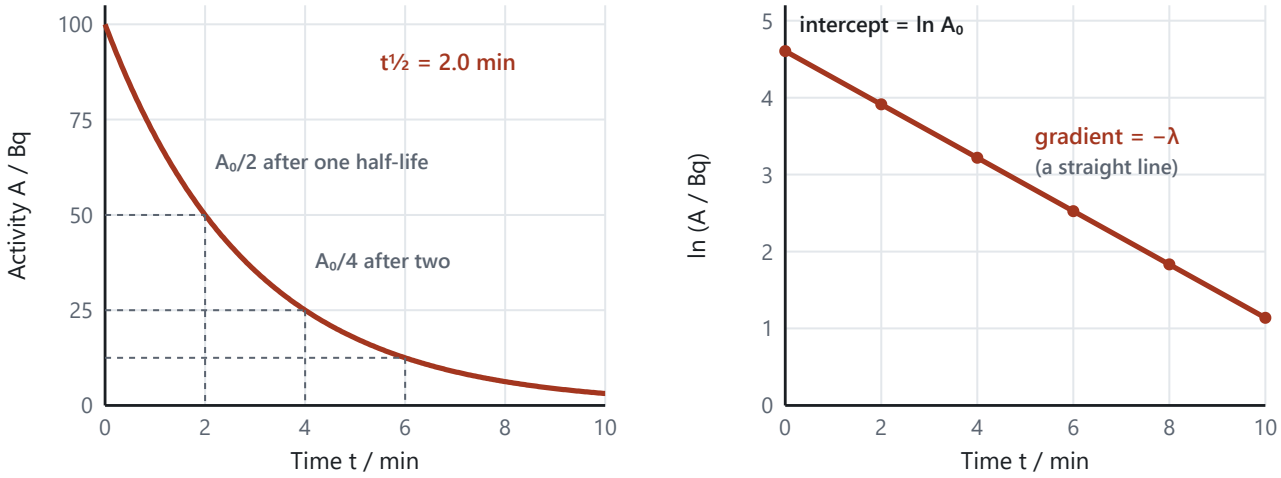
142 The log forms

- Take natural logs of $A = A_0e^{-\lambda t}$:

$$\ln A = \ln A_0 - \lambda t$$

- Compare with $y = mx + c$: plotting **ln A against t** gives a straight line with **gradient −λ** and intercept $\ln A_0$. The same works for N: $\ln N = \ln N_0 - \lambda t$.
- Rearranged for time:

$$t = (1/\lambda)\ln(A_0/A)$$



Left: activity halves every half-life (here 2.0 min), and the curve never reaches zero. Right: ln A against t is a straight line of gradient −λ, which is the best way to find λ from data.

142 Finding the half-life from a graph

- **From an A–t (or N–t) curve**: read A_0, find when $A = A_0/2$; check with $A_0/4$ and $A_0/8$ (they should come at $2t_{½}$ and $3t_{½}$) and average.
- **From a ln A–t line**: gradient = −λ, then $t_{½} = \ln 2/\lambda$. Use a large triangle.
- Subtract **background** from count rates first.

142 Modelling decay step by step

The rate equation $\Delta N/\Delta t = -\lambda N$ can be modelled on a spreadsheet: choose a small Δt, then repeatedly calculate $\Delta N = -\lambda N\Delta t$ and $N_{new} = N + \Delta N$. The smaller Δt is, the closer the result to the exponential curve.

WORKED EXAMPLE: IODINE-131 ($T_{½}$ = 8.0 DAYS)

A sample has activity 4.0×10^8 Bq.
(a) $\lambda = \ln 2/t_{½} = 0.693/(8.0 \times 24 \times 3600\ s) =$ **$1.0 \times 10^{-6}\ s^{-1}$**
(b) $N = A/\lambda = 4.0 \times 10^8 / 1.0 \times 10^{-6} =$ **4.0×10^{14} nuclei**
(c) Activity after 30 days ($\lambda = 0.693/8.0 = 0.0866\ day^{-1}$, so t can stay in days):
$A = 4.0 \times 10^8 \times e^{-0.0866 \times 30} = 4.0 \times 10^8 \times 0.0743 =$ **3.0×10^7 Bq**
(d) Time to fall to 1.0% of A_0:
$t = (1/\lambda) \ln(A_0/A) = (1/0.0866) \times \ln(100) = 4.61/0.0866 =$ **53 days**

WORKED EXAMPLE: ACTIVITY FROM A MASS

Find the activity of 5.0 mg of cobalt-60 ($t_{½}$ = 5.27 years; 1 year = 3.15×10^7 s).
N = mass/mass of one atom = 5.0×10^{-6} kg / ($60 \times 1.66 \times 10^{-27}$ kg) = 5.02×10^{19}
$\lambda = 0.693/(5.27 \times 3.15 \times 10^7\ s) = 4.18 \times 10^{-9}\ s^{-1}$
$A = \lambda N = 4.18 \times 10^{-9} \times 5.02 \times 10^{19} =$ **2.1×10^{11} Bq**

WORKED EXAMPLE: CARBON DATING

Living wood has a C-14 activity of 0.25 Bq per gram of carbon. Ancient wood gives 0.21 Bq per gram. $t_{½}$ of C-14 = 5730 years. How old is it?
$\lambda = 0.693/5730 = 1.21 \times 10^{-4}\ y^{-1}$
$t = (1/\lambda) \ln(A_0/A) = \ln(0.25/0.21)/1.21 \times 10^{-4} = 0.174/1.21 \times 10^{-4} =$ **1400 years**

EXAM TIP

Decide first: does the question want t, A or N? Then choose units: if you need $A = \lambda N$ (so A in Bq), λ **must** be in s^{-1}; if you only use $A = A_0e^{-\lambda t}$, any matching units work. Keep 3–4 s.f. in λ, especially when A and A_0 are close, and finish with the comparison if there's a claim.

MARK-SCHEME LANGUAGE

- "Use of $\lambda = \ln 2/t_{½}$", "use of $A = \lambda N$", "use of $A = A_0e^{-\lambda t}$", "conversion between seconds and years" (its own mark) (WPH15 Oct 2025).
- "Allow ecf for λ." For deduce questions the final mark is the comment: "X > Y so not consistent", or "X ≈ Y so consistent" (WPH15 Jun 2026).
- Power from a source: P = A × (energy per decay in J) (WPH15 Jun 2021).

Often asked as: find N or N_0; time for the activity to fall to a level; activity after a time, "is the claim true?"; compare two isotopes; power from a radioactive source; percentage composition from activity; MCQ calculating λ in s^{-1}.

COMMON MISTAKE

Putting the **half-life** into the exponent instead of the elapsed time. Mixing up N and N_0, or A and A_0 (a measured "2.35 GBq" was A, not A_0). Not converting days or years to seconds before $A = \lambda N$. Treating "0.75%" as 0.75 instead of 0.0075. Using $(½)^n$ with a non-whole n without logs. Concluding "the claim is accurate" when your number disagrees, even at 1 s.f.

BEYOND THE SPEC, BUT EXAMINED

- Number of nuclei from a sample's mass and the atomic mass unit (WPH15 Oct 2025, Jun 2026).
- When the time is much shorter than the half-life, the activity is nearly constant over it (WPH15 Oct 2023).
- Ratio questions: for two isotopes with the same activity, $N_1/N_2 = \lambda_2/\lambda_1 = t_{1/2_1}/t_{1/2_2}$ (WPH15 Oct 2024).

Topic 5.5 – Oscillations

SPEC 143 The condition for simple harmonic motion

WHAT THIS SPEC POINT ASKS

Understand that SHM needs a restoring force F = −kx, and use this to decide when SHM will happen.

143 Definition

An object moves with **simple harmonic motion** (SHM) when its **acceleration (or resultant force) is proportional to its displacement from the equilibrium position, and always acts in the opposite direction to the displacement** (towards the equilibrium position).

$$F = -kx \quad \text{so} \quad a = -(k/m)x$$

- x = displacement from the equilibrium position (m); k = a constant (for a spring, the stiffness, N m^{-1}).
- The **minus sign** means the force points the opposite way to the displacement: it is a **restoring** force.
- Since a = −(k/m)x and k/m is constant, comparing with a = $-\omega^2 x$ (144) gives **$\omega^2 = k/m$**.

143 Spotting SHM

System	Restoring force	SHM?
Mass on a spring (within the limit of proportionality)	Hooke's law: F = −kx	yes
Simple pendulum, **small** angles (under about 10°)	F ≈ −(mg/l)x	approximately
A ball bouncing on the floor	weight, constant while in the air (not ∝ x)	no
Pendulum swung through a large angle	not proportional to x	no
Liquid oscillating in a U-tube; a floating object bobbing	∝ displacement	yes

- The test is always: write the resultant force (or acceleration) in terms of x. If it is "−(constant) × x", the motion is SHM.

WORKED EXAMPLE: RESTORING FORCE AND ACCELERATION

A 0.40 kg mass on a spring (k = 25 N m^{-1}) is pulled 3.0 cm below its equilibrium position and released. Find the force and acceleration at release.

F = −kx = −25 × 0.030 = −0.75 N (magnitude 0.75 N, upwards, towards equilibrium)

a = F/m = −0.75/0.40 = **−1.9 m s^{-2}** (1.9 m s^{-2} upwards)

MARK-SCHEME LANGUAGE

Two marks, asked in 15 of 18 papers:

- "(resultant) force / acceleration is **proportional to the displacement** (from the equilibrium position)";
- "and acts in the **opposite direction to the displacement**" (or "(always) towards the equilibrium position").
- "Equilibrium **position**" (or "undisplaced / fixed / central point or position") is needed: "reject 'equilibrium'" on its own (WPH15 Jun 2026); "do not accept mean position" (WPH15 Oct 2022). One paper rejected "towards the equilibrium position" and wanted "opposite to the displacement" (WPH15 Oct 2025): use that wording to be safe.
- "An equation with symbols defined correctly is a valid response for both marks", but every term must be defined and the minus sign explained (WPH15 Jun 2023).

Often asked as: state the condition for SHM; explain how an a–x graph shows SHM; decide which situation is SHM.

COMMON MISTAKE

Writing "equilibrium" without "position". Saying "acceleration is proportional to displacement" and stopping (no direction). Writing $a = -\omega^2 x$ without saying what a, ω and x are or what the minus sign means. Repeating the learnt definition when the question says "explain how the **graph** shows SHM": refer to the graph (a straight line through the origin with a negative gradient).

BEYOND THE SPEC, BUT EXAMINED

- For an a–x graph question: "straight line **through the origin**" gives proportionality and "**negative gradient**" gives the opposite direction; the gradient is $-\omega^2$ (WPH15 Jan 2025).

SPEC 144

The SHM equations

WHAT THIS SPEC POINT ASKS

Use $a = -\omega^2 x$, $x = A\cos\omega t$, $v = -A\omega\sin\omega t$, $a = -A\omega^2\cos\omega t$, $T = 1/f = 2\pi/\omega$ and $\omega = 2\pi f$ for a simple harmonic oscillator.

144 The equations

$$a = -\omega^2 x$$

$$x = A\cos\omega t \qquad v = -A\omega\sin\omega t \qquad a = -A\omega^2\cos\omega t$$

$$\omega = 2\pi f = 2\pi/T \qquad T = 1/f$$

- A = **amplitude** (maximum displacement, m); ω = **angular frequency** (rad s^{-1}); f = frequency (Hz); T = period (s); t = time (s).
- $x = A\cos\omega t$ assumes the timing starts at maximum displacement (released from x = A). If it starts at the equilibrium position moving in the +x direction, $x = A\sin\omega t$ instead.
- v is the gradient of x–t, and a is the gradient of v–t (146, 147), which is why each line follows from the one before.
- Substituting $x = A\cos\omega t$ into $a = -A\omega^2\cos\omega t$ gives $a = -\omega^2 x$, the SHM condition.

144 Maximum values

Quantity	Maximum	Where
Speed	$v_{max} = A\omega$ (when $\sin \omega t = \pm 1$)	at the **equilibrium position** ($x = 0$)
Acceleration	$a_{max} = A\omega^2$ (when $\cos \omega t = \pm 1$)	at the **extremes** ($x = \pm A$)

- The period of SHM **does not depend on the amplitude** (it is isochronous).

WORKED EXAMPLE: AMPLITUDE AND MAXIMUM ACCELERATION

A loudspeaker cone vibrates in SHM at 240 Hz with a maximum speed of 2.25 m s^{-1}.
$\omega = 2\pi f = 2\pi \times 240 = 1508$ rad s^{-1}
$A = v_{max}/\omega = 2.25/1508 = 1.49 \times 10^{-3}$ m = **1.49 mm**
$a_{max} = A\omega^2 = 1.49 \times 10^{-3} \times 1508^2 =$ **3.39×10^3 m s^{-2}**

WORKED EXAMPLE: DISPLACEMENT AND VELOCITY AT A GIVEN TIME

A mass oscillates with A = 4.0 cm and T = 0.80 s, released from $x = +A$ at $t = 0$. Find x and v at $t = 0.30$ s.
$\omega = 2\pi/T = 2\pi/0.80 = 7.85$ rad s^{-1}; $\omega t = 7.85 \times 0.30 = 2.36$ rad (calculator in **radians**)
$x = A \cos \omega t = 4.0 \times \cos 2.36 =$ **−2.8 cm** (2.8 cm on the other side of equilibrium)
$v = -A\omega \sin \omega t = -0.040 \times 7.85 \times \sin 2.36 =$ **−0.22 m s^{-1}** (moving in the −x direction)

EXAM TIP

Convert first: frequency "per minute" to Hz, cm or mm to m. Use **radians** for ωt. For maxima you don't need t at all: $v_{max} = A\omega$, $a_{max} = A\omega^2$.

MARK-SCHEME LANGUAGE

- "Use of $\omega = 2\pi/T$" (or $2\pi f$) is its own mark; "use of $v = A\omega \sin \omega t$ with $\sin \omega t = 1$".
- Maximum KE: "use of $E_k = \frac{1}{2}mv^2$" with $v_{max} = A\omega$; or $E_{k,max} = \Delta E_{el} = \frac{1}{2}kA^2$ "because energy is conserved" (WPH15 Oct 2020).

Often asked as: amplitude or v_{max} from f; heart-beat rate per minute to Hz then amplitude; maximum KE from T and A; reading v_{max} from a graph; MCQs on $v_{max} = 2\pi A/T$ and phase.

COMMON MISTAKE

Calculator in degrees. Confusing f and T (or f and ω). Not seeing that maximum speed means $\sin \omega t = 1$. Giving an answer in m when mm was asked. Not squaring 2π when rearranging.

BEYOND THE SPEC, BUT EXAMINED

- Speed at any displacement: **$v = \pm\omega\sqrt{(A^2 - x^2)}$**, which comes from energy conservation (150). Useful when you know x but not t.
- Acceleration is in **antiphase** with displacement (half a cycle out); velocity is a quarter cycle out of phase with both (WPH15 Jan 2021).

SPEC 145 Periods of a mass-spring system and a simple pendulum

WHAT THIS SPEC POINT ASKS

Use $T = 2\pi\sqrt{(m/k)}$ for a mass on a spring and $T = 2\pi\sqrt{(l/g)}$ for a simple pendulum.

145 Mass on a spring

$$T = 2\pi\sqrt{(m/k)}$$

- m = oscillating mass (kg); k = spring constant (N m^{-1}). From $\omega^2 = k/m$ and $T = 2\pi/\omega$.
- Heavier mass → longer period; stiffer spring → shorter period. $T \propto \sqrt{m}$: four times the mass doubles T.
- The same T applies horizontally or vertically (gravity only shifts the equilibrium position).
- Find k from a static test: $k = F/\Delta x = mg/\Delta x$.

145 Simple pendulum (small angles)

$$T = 2\pi\sqrt{(l/g)}$$

- l = length from the pivot to the **centre of mass** of the bob (m); g = 9.81 N kg^{-1}.
- T depends only on **l and g**: **not** on the mass of the bob and not on the amplitude (as long as the angle is small, under about 10°).
- For a pendulum, $\omega^2 = g/l$.

145 Getting k or g from a straight-line graph

Plot	Straight line because	Gradient gives
T^2 against m	$T^2 = (4\pi^2/k)\,m$	$k = 4\pi^2/\text{gradient}$
T^2 against l	$T^2 = (4\pi^2/g)\,l$	$g = 4\pi^2/\text{gradient}$
f^2 against 1/m	$f^2 = (k/4\pi^2)(1/m)$	$k = 4\pi^2 \times \text{gradient}$

WORKED EXAMPLE: A SECONDS PENDULUM

Find the length of a pendulum with T = 2.000 s (g = 9.81 N kg^{-1}).
$l = gT^2/4\pi^2 = 9.81 \times 2.000^2/(4\pi^2) =$ **0.994 m**
It can't be used to define the metre exactly: **g varies with location**.

WORKED EXAMPLE: EXTRA MASS ON A SPRING

A 0.250 kg mass on a spring has T = 0.60 s. What mass must be added to make T = 0.75 s?
$k = 4\pi^2 m/T^2 = 4\pi^2 \times 0.250/0.60^2 = 27.4$ N m^{-1}
New total mass $m = kT^2/4\pi^2 = 27.4 \times 0.75^2/(4\pi^2) = 0.391$ kg
Added mass = 0.391 − 0.250 = **0.14 kg**
(Ratio shortcut: $m_2 = m_1(T_2/T_1)^2 = 0.250 \times (0.75/0.60)^2 = 0.391$ kg.)

EXAM TIP

Substitute numbers first, then rearrange: most lost marks come from dropping the 2π or the square root while rearranging symbols. Time **many** oscillations (e.g. 20 or 30) and divide, rather than timing one.

MARK-SCHEME LANGUAGE

- "Use of $T = 2\pi\sqrt{(m/k)}$" with the **total** oscillating mass; "allow use of $\omega^2 = k/m$ with $T = 2\pi/\omega$".
- "Use of $T = t_{mean}/30$" for repeated timing.
- Two springs side by side sharing the load: "factor of 2 applied to either m or k" (WPH15 Jun 2023).
- Pendulum period doesn't depend on mass: adding mass to the bob changes nothing (except through l) (WPH15 Jan 2025).

Often asked as: pendulum length for a given T; pendulum on another planet (find g from $g = GM/r^2$ first); k from a load test then f; mass added to change T; k from an f^2 against 1/m graph.

COMMON MISTAKE

Stopping at the total mass when the question asks for the mass **added**. Forgetting the factor 2 for two springs. Using orbital equations when the question is about a **pendulum** on a planet (it scored zero). Rearranging before substituting.

BEYOND THE SPEC, BUT EXAMINED

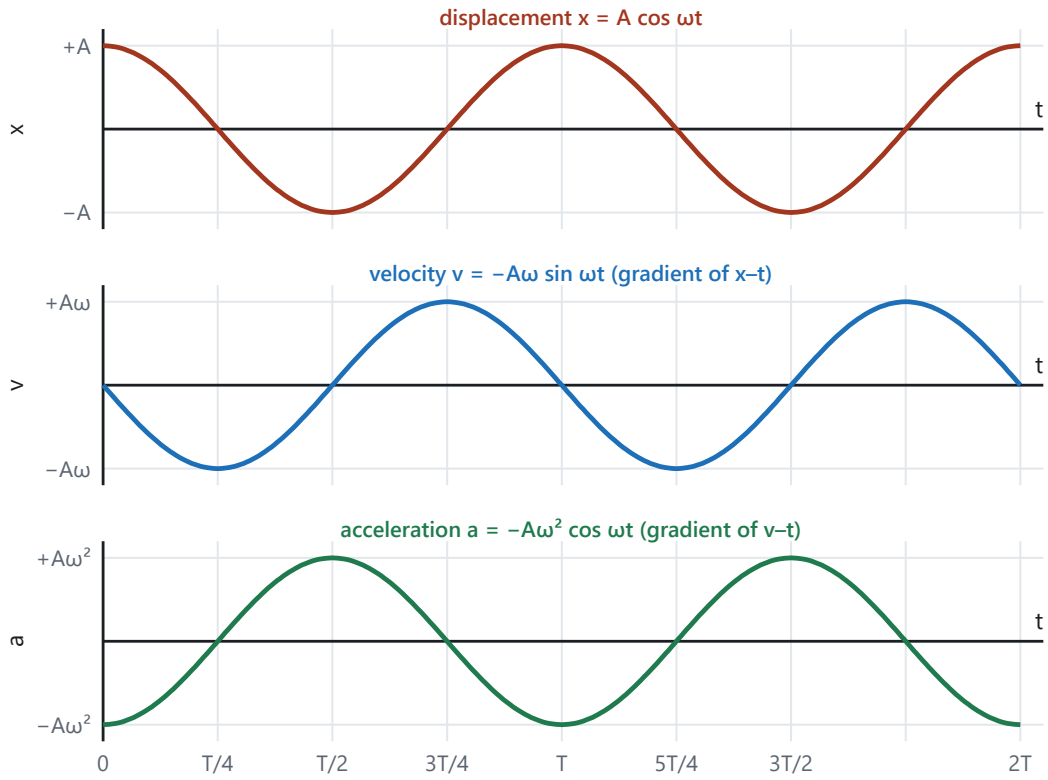
- **Springs in parallel** share the load, so the effective k doubles; in series the effective k halves (WPH15 Jun 2023).
- The maximum load a spring can take is smaller when oscillating than at rest, because at the lowest point the spring force exceeds the weight (WPH15 Oct 2021).

SPEC 146

Displacement-time graphs

WHAT THIS SPEC POINT ASKS

Draw and interpret a displacement-time graph for an oscillating object, knowing that its gradient at any point gives the velocity.



SHM starting from x = +A. Each graph is the gradient of the one above. v is a quarter cycle out of phase with x; a is exactly opposite to x (antiphase).

146 Reading the x–t graph

- **Amplitude** A = the height of a peak above the equilibrium line.
- **Period** T = time for one complete cycle (peak to next peak). Read it over several cycles and divide, for accuracy. Then f = 1/T and ω = 2π/T.
- **Gradient = velocity**:
  - at a **peak or trough** the gradient is **zero**: the object is momentarily at rest at the extreme;
  - where the graph **crosses the time axis** the gradient is **steepest**: maximum speed, at the equilibrium position.
- To find v at a point, draw a tangent and find its gradient (use a large triangle).

146 Sketching

- Use a sine or cosine shape depending on where timing starts: x = A cos ωt from an extreme, x = A sin ωt from the equilibrium position, x = −A sin ωt if it starts at equilibrium moving in the −x direction.
- Keep the **amplitude constant** (if undamped) and the period the same as any other graph given. Put **numbers on the time axis** (e.g. T/2, T or the actual times).

WORKED EXAMPLE: MAXIMUM SPEED FROM A DISPLACEMENT-TIME GRAPH

An x–t graph shows A = 3.0 cm and three complete cycles in 1.20 s.

T = 1.20/3 = 0.40 s

$v_{max} = A\omega = 2\pi A/T = 2\pi \times 0.030/0.40 =$ **0.47 m s^{-1}**

Check: the tangent where the graph crosses the axis should have a gradient of about 0.47 m s^{-1} (answers 0.45–0.55 m s^{-1} were accepted from a tangent) (WPH15 Jun 2025).

MARK-SCHEME LANGUAGE

- Sketch from a v–t graph: "same time period as velocity and constant amplitude"; "wave shifted a quarter cycle" (WPH15 Jun 2021).
- "Minus sine graph", "approximately the same period", amplitude approximately constant (WPH15 Oct 2025).

Often asked as: sketch x–t to match a given v–t or a–t graph; read A and T; v_{max} from the gradient at the axis crossing.

COMMON MISTAKE

Leaving the time axis without values. Drawing the amplitude changing when there is no damping. Shifting by half a cycle instead of a quarter. Timing from one cycle only.

SPEC 147 Velocity-time graphs

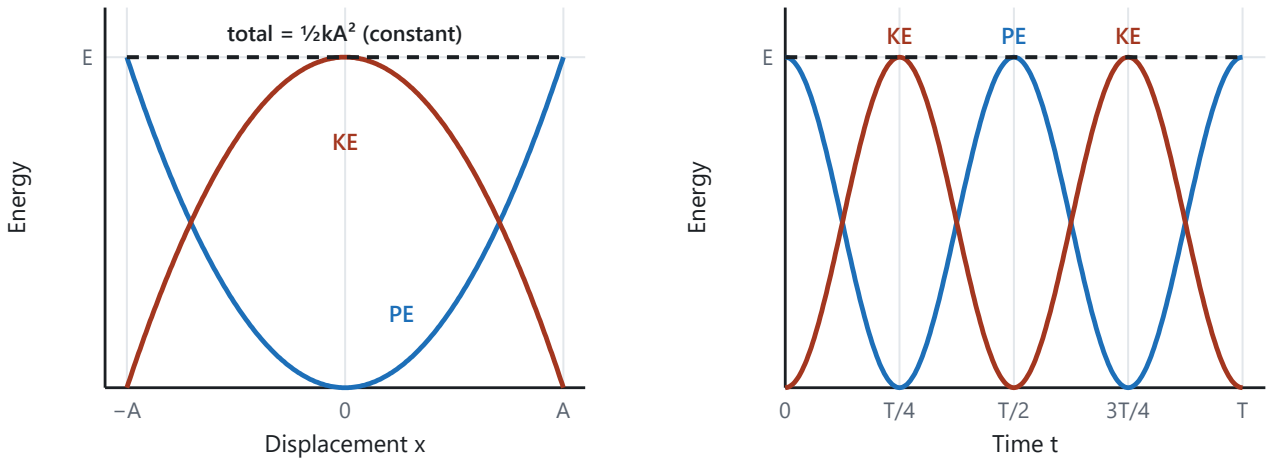
WHAT THIS SPEC POINT ASKS

Draw and interpret a velocity-time graph for an oscillating object, knowing that its gradient at any point gives the acceleration.

147 Reading the v–t graph

- The v–t graph is the gradient of the x–t graph: for $x = A\cos\omega t$ it is $v = -A\omega\sin\omega t$, a quarter cycle out of phase with x. Its peaks are at $\pm A\omega$.
- **Gradient = acceleration**:
  - where v crosses zero (object at an **extreme**) the v–t gradient is **steepest**: maximum acceleration, $\pm A\omega^2$;
  - at a **peak** of v (object at **equilibrium**) the gradient is **zero**: no acceleration, because there is no resultant force at x = 0.
- The **area** under a v–t graph from one extreme to the next gives the distance travelled, 2A.
- The a–t graph (gradient of v–t) is $a = -A\omega^2\cos\omega t$: the mirror image of the x–t graph.

147 Energy-time graphs



Energy in undamped SHM. Against displacement (left), PE is a ∪-shaped parabola, KE an ∩-shaped one, and they add to a constant total. Against time (right), each energy is always positive and goes through two cycles for every oscillation.

- KE = ½mv², so KE is always **positive** and peaks every time the object passes through equilibrium, which happens **twice per cycle**. Energy–time graphs have **twice the frequency** of the displacement.

MARK-SCHEME LANGUAGE

- Kinetic energy against time: "positive", "frequency double that of the displacement", "maximum at t = 0 when the mass passes the equilibrium position" (WPH15 Oct 2020).
- Energy against displacement is a **parabola** (WPH15 Jan 2025).

Often asked as: MCQ on the a–t graph from the gradient of a v–t graph; sketch KE against time; sketch energy against displacement.

COMMON MISTAKE

For energy against displacement, drawing a sine curve, a straight line, an upside-down parabola for PE, or negative energies. For KE against time, drawing it at the same frequency as x.

SPEC 148 Resonance

WHAT THIS SPEC POINT ASKS

Understand what resonance is.

148 The idea

- Every oscillating system has a **natural frequency** f_0: the frequency it oscillates at when displaced and left alone.
- When a periodic **driving force** acts at a frequency equal to the natural frequency, energy is transferred to the system most efficiently (each push comes at the same point in each cycle, in step with the motion). The amplitude builds up to a **maximum**.

Resonance: when the **driving (forcing) frequency equals the natural frequency** of the system, there is **maximum transfer of energy** to the system and the amplitude of oscillation is a **maximum**.

148 Examples

Useful	Harmful (must be damped or avoided)
A child on a swing pushed once per swing	A bridge driven by people walking or by wind (the London Millennium Bridge)
Rocking a car stuck in mud, in time with its natural rocking	Buildings in an earthquake whose natural frequency matches the ground's shaking
Musical instruments: air columns and strings resonate	Car parts or panels rattling at a particular engine speed
Tuning a radio circuit to one station's frequency	A wine glass shattered by a sound at its natural frequency

- **Barton's pendulums**: a heavy driver pendulum hangs from a string with several light pendulums of different lengths. The one with the **same length** as the driver (same natural frequency) swings with the largest amplitude.

MARK-SCHEME LANGUAGE

- "The **driving (forcing) frequency equals the natural frequency** of [the system named]", "**maximum transfer of energy**" (accept "efficient transfer"), "resulting in maximum amplitude".
- Name both frequencies for the system in the question: "driving frequency of the hand equals the natural frequency of the mass-spring system" (WPH15 Jun 2023: very few did).
- Wine glass (6-mark) (WPH15 Jun 2022): striking gives a **free** oscillation that dies away (damping); a wet finger sliding round the rim **drives / forces** the glass; resonance when the driving frequency equals (is close to) the natural frequency; large amplitude, so loud sound.
- Bridge (6-mark) (WPH15 Jan 2025): people force the bridge to oscillate; "large amplitude of oscillation" (not "large swaying"); energy transferred from the bridge to the dampers.

Often asked as: define resonance in a context (tide in a bay, a hand driving a spring, an animal on a spring balance, a carriage on a bumpy road); decide whether a situation is resonance (microwave at 2.45 GHz against a natural frequency near 10 GHz: no); bridge and building design.

COMMON MISTAKE

Calling every oscillation resonance: striking a glass is a free oscillation, not resonance. Saying "the frequencies match" without saying **which** frequencies, of **what**. Writing only "maximum amplitude" when the mark needs "maximum energy transfer". Assuming resonance is always bad.

BEYOND THE SPEC, BUT EXAMINED

- Resonance can still occur when the driver pushes at a **sub-multiple** of the natural frequency (every second or third cycle), because each push still comes at the same point in the cycle (WPH15 Jan 2021: insect wing muscles).
- Not every "frequency" heating is resonance: a microwave oven runs at 2.45 GHz, well away from the natural frequency quoted for water molecules (near 10 GHz), so the answer to "is this resonance?" was **no** (WPH15 Oct 2020). Check the numbers before saying resonance.
- At resonance the energy supplied per cycle equals the energy lost to damping per cycle, so the amplitude stops growing (WPH15 Jun 2023).

SPEC 149 Core practical 16: unknown mass from resonant frequencies

WHAT THIS SPEC POINT ASKS

Find an unknown mass by measuring the resonant frequencies of a spring loaded with known masses.

149 Principle

- A mass m on a spring has natural (resonant) frequency $f = 1/T = (1/2\pi)\sqrt{(k/m)}$, so

$$f^2 = (k/4\pi^2) \times (1/m) \quad \text{or} \quad T^2 = (4\pi^2/k) \times m$$

- Measure f for several known masses, plot a calibration graph (f^2 against 1/m, or T^2 against m), then measure f for the unknown mass and read its mass off the graph.

CORE PRACTICAL 16: METHOD

1. Hang a spring from a clamp. Attach its bottom end to a known mass, and connect a **vibration generator** driven by a **signal generator** to the top of the spring (or below the mass).

2. Slowly increase the frequency from low values and watch the amplitude. Find the frequency at which the amplitude is **largest**: the **resonant frequency**. Approach it from above and below and average; use the signal generator's display or an oscilloscope to read f.
3. Repeat for 5–6 known masses (e.g. 100 g to 600 g).
4. Plot f^2 (y) against 1/m (x): a straight line through the origin with gradient $k/4\pi^2$.
5. Hang the unknown mass, find its resonant frequency, and read 1/m from the graph (or calculate m = gradient/f^2).

Alternative: displace each mass and let it oscillate freely. The natural frequency equals the resonant frequency, so timing 20 oscillations (T = t/20, f = 1/T) gives the same data. Plot T^2 against m.
Variables: independent = mass; dependent = resonant frequency (or period); control = same spring, small amplitude, same clamp.
Safety: clamp the stand to the bench (or weight the base); wear eye protection in case the spring snaps; keep feet clear of falling masses.

WORKED EXAMPLE: READING THE UNKNOWN MASS

A T^2–m graph for a spring is a straight line through the origin with gradient 1.97 s^2 kg^{-1}. The unknown mass oscillates with T = 0.85 s.
$k = 4\pi^2/\text{gradient} = 39.5/1.97 = 20$ N m^{-1}
$m = T^2/\text{gradient} = 0.85^2/1.97 =$ **0.37 kg**

149 Errors and improvements

- Resonance peaks are hard to judge by eye: find the frequency range where the amplitude is largest, take its midpoint, and repeat.
- The spring's own mass adds to the oscillating mass, so a T^2–m line may not pass exactly through the origin. That doesn't stop the method working, because the unknown is read off the same line.
- Keep amplitudes small so the spring obeys Hooke's law.
- For timing: start and stop at the **equilibrium position** (fastest point, using a fiducial marker); time many oscillations.

MARK-SCHEME LANGUAGE

- "Combine $T = 2\pi\sqrt{(m/k)}$ with f = 1/T to obtain $f^2 = (k/4\pi^2)m^{-1}$", "compare with y = mx + c", "large triangle used", "conclusion consistent with the earlier value" (WPH15 Jun 2021).
- "Use of T = t/30" when 30 oscillations were timed (WPH15 Jun 2023).

Often asked as: the procedure itself hasn't been asked yet; the f^2 against 1/m analysis and repeated timing have. A data logger's advantage: many readings over very short or very long times (WPH15 Jun 2021).

COMMON MISTAKE

Reading T from a single oscillation. Plotting f against m and expecting a straight line. Forgetting that the gradient of f^2 against 1/m is $k/4\pi^2$, not k.

SPEC 150 Energy in undamped and damped oscillations

WHAT THIS SPEC POINT ASKS

Apply conservation of energy to undamped and damped oscillating systems.

150 Undamped: energy swaps between KE and PE

- With no energy losses, the total energy is **constant**. At the extremes it is all **potential** (elastic PE for a spring, gravitational PE for a pendulum); at the equilibrium position it is all **kinetic**.
- For a mass on a spring (taking PE = 0 at equilibrium):

$$E_{total} = \tfrac{1}{2}kA^2 = \tfrac{1}{2}mv_{max}^2$$

$$\text{at displacement } x: \quad \tfrac{1}{2}mv^2 + \tfrac{1}{2}kx^2 = \tfrac{1}{2}kA^2$$

- **Energy ∝ amplitude²**. Double the amplitude and the energy goes up four times; halve the energy and the amplitude falls to 1/√2 (71%).

WORKED EXAMPLE: SPEED FROM ENERGY

A 0.20 kg mass on a spring (k = 50 N m^{-1}) oscillates with amplitude 4.0 cm. Find the total energy, v_{max}, and the speed at x = 2.0 cm.

$E = \tfrac{1}{2}kA^2 = \tfrac{1}{2} \times 50 \times 0.040^2 =$ **0.040 J**

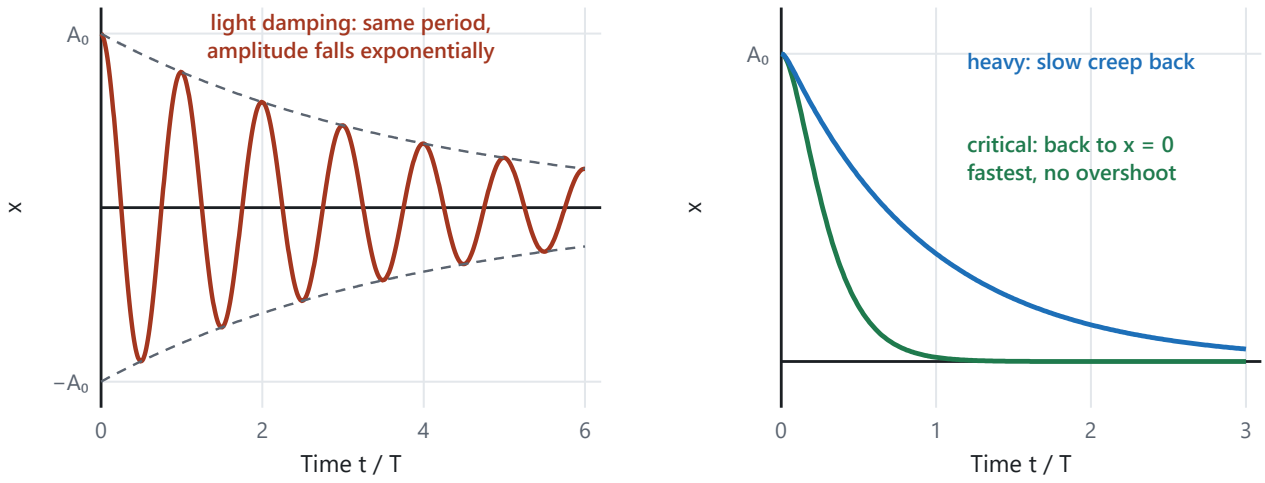
$v_{max} = \sqrt{(2E/m)} = \sqrt{(2 \times 0.040/0.20)} =$ **0.63 m s^{-1}** (check: $\omega = \sqrt{(k/m)} = 15.8$ rad s^{-1}, $A\omega = 0.63$ m s^{-1})

At x = 0.020 m: PE = ½ × 50 × 0.020^2 = 0.010 J, so KE = 0.040 − 0.010 = 0.030 J

$v = \sqrt{(2 \times 0.030/0.20)} =$ **0.55 m s^{-1}**

150 Damped: energy leaves the system

- **Damping**: **resistive forces** (air resistance, friction, a liquid's drag) act on the oscillator. **Work is done against** them, so energy is **transferred out of the oscillating system** to the **surroundings** (their thermal energy store rises).
- The total energy falls, so the **amplitude falls** with time. With light damping the period stays almost the same.
- The damping force always acts **opposite to the velocity** (it is not the restoring force, which acts opposite to the displacement).



Left: light damping. The period is unchanged; the amplitude falls by the same fraction each cycle (an exponential decay). Right: critical damping returns to equilibrium in the shortest time without overshooting; heavy damping returns slowly without oscillating.

Degree of damping	Behaviour	Example
Light	oscillates, amplitude decays gradually	a pendulum in air
Critical	returns to equilibrium in the shortest time, no oscillation	car suspension, a measuring needle
Heavy	returns very slowly, no oscillation	a door closer

MARK-SCHEME LANGUAGE

- Damping: "energy is transferred (dissipated) to the surroundings" (or "the thermal energy store of the dampers increases") "because **work is done against resistive forces**" (WPH15 Jun 2024, Jun 2025, Jun 2026). Only "MAX 1 for reference to damping" without the work-done idea (WPH15 Oct 2023).
- "Work is done (by the roller) as oil is forced through the holes" (damper in a building) (WPH15 Jun 2024).
- Undamped: "elastic PE converted to KE at the equilibrium position, total energy constant".

Often asked as: maximum KE from elastic PE; draw a horizontal total-energy line on an energy–displacement graph and read off energies; why a mass oscillating in water stops; MCQs on amplitude $\propto \sqrt{E}$ and on the direction of the damping force.

COMMON MISTAKE

Explaining damping without "work done against resistive forces". Confusing the damping force with the resultant (restoring) force. Saying KE is constant, or zero at equilibrium.

BEYOND THE SPEC, BUT EXAMINED

- Total energy = $\frac{1}{2}kA^2$, the area under the force–extension graph up to the amplitude (WPH15 Oct 2020, Jan 2023).
- The damping force is always opposite to the **velocity** (WPH15 Oct 2023, Oct 2025 MCQs).

SPEC 151

Free and forced oscillations

WHAT THIS SPEC POINT ASKS

Distinguish free oscillations from forced oscillations.

151 The difference

	Free oscillation	Forced oscillation
How it starts	displaced once (or struck), then left alone	a **periodic external (driving) force** keeps acting
Frequency	the system's **natural frequency** f_0	the **driving frequency** (once it has settled)
Energy	no energy supplied after the start; with damping, the amplitude dies away	energy is supplied by the driver each cycle, replacing the energy lost to damping
Examples	a plucked guitar string; a struck tuning fork; a pendulum pulled aside and released	a swing pushed every cycle; a car body vibrating as it drives over a bumpy road; the cone of a loudspeaker

- A forced oscillator first goes through a short messy stage, then settles to a steady amplitude at the driving frequency. How big that amplitude is depends on how close the driving frequency is to f_0 (spec point 152).

MARK-SCHEME LANGUAGE

- Free: oscillates "at its natural frequency" after "a single displacement", with no external periodic force.
- Forced: "an external periodic force" makes it oscillate "at the driving frequency".
- Pushing a swing each time it reaches maximum height is a **forced** oscillation (WPH15 Oct 2025).

Often asked as: classify situations (MCQ); the striking/rubbing wine-glass 6-mark question (WPH15 Jun 2022).

COMMON MISTAKE

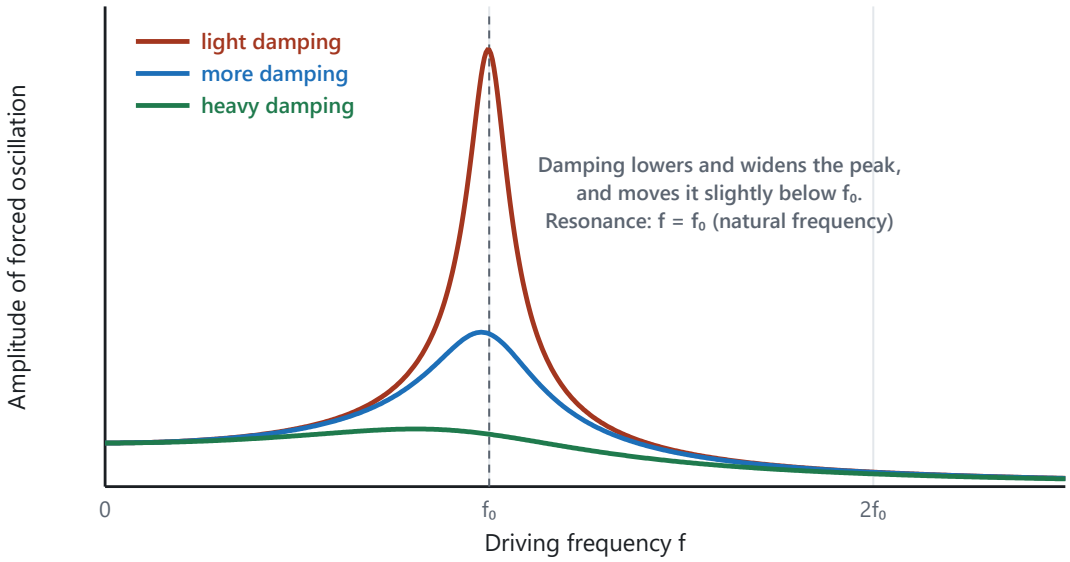
Describing both the struck glass and the rubbed glass as resonance: striking is free; rubbing is forced (and can produce resonance). Defining "forced" as "not free" without mentioning the external periodic force or the energy it supplies.

SPEC 152 Forced oscillations near the natural frequency; damping and resonance

WHAT THIS SPEC POINT ASKS

Understand how the amplitude of a forced oscillation varies with driving frequency, especially at and near the natural frequency, and know qualitatively how damping changes resonance.

152 The resonance curve



Amplitude of a forced oscillation against driving frequency. The peak is at (or just below) the natural frequency f_0. More damping gives a lower, broader peak, shifted slightly to lower frequency.

- **Low driving frequency**: the system just follows the driver; amplitude ≈ the driver's own amplitude (the static displacement).
- **As f approaches f_0**: amplitude rises sharply; maximum at resonance (f ≈ f_0).
- **Above f_0**: amplitude falls, towards zero at very high frequency (the system can't keep up).

152 Effect of damping

- More damping means more energy is removed each cycle, at **every** frequency, so:
  - the **peak amplitude is lower**;
  - the peak is **broader** (flatter, less sharp), so the system responds over a wider range of frequencies;
  - the peak moves **slightly below** f_0.
- Heavy damping can remove the peak almost entirely.
- Damping **doesn't change the natural frequency** itself (very little, for light damping).

152 Why it matters

- Buildings in earthquake zones and long bridges are fitted with **dampers** (oil-filled pistons, tuned mass dampers, rubber bearings) so that if they are driven near their natural frequency the amplitude stays small. The damper is designed so that **energy is transferred efficiently into the damper**, where work done against resistive forces dissipates it.
- Where a sharp resonance is wanted (radio tuning, musical instruments), damping is kept small.

MARK-SCHEME LANGUAGE

- "Damping reduces the amplitude (of the peak)" and "the peak is flatter / broader"; energy is removed "at all frequencies" (WPH15 Jun 2021).
- Damper design: "maximum energy transfer to the damper" (from the building) (WPH15 Jun 2024).

Often asked as: MCQs on where damping acts; designing dampers for a building or carriage; describe how amplitude changes as the driving frequency is increased. A sketch of the curves hasn't been asked in recent papers, but learn it.

COMMON MISTAKE

Saying damping changes the natural frequency a lot, or moves the peak to a higher frequency. Drawing heavier-damping curves with taller peaks. Designing a building damper to "avoid resonance" when the question wants energy transferred into the damper.

SPEC 153 Damping by plastic deformation of ductile materials

WHAT THIS SPEC POINT ASKS

Understand how damping and the plastic deformation of ductile materials reduce the amplitude of oscillations.

153 Ductile materials

- A **ductile** material can undergo **large plastic deformation** (it can be drawn into wires) before it breaks. Examples: copper, mild steel, gold.
- **Plastic** deformation is **permanent**: the material doesn't return to its original shape when the force is removed.

153 How they damp oscillations

- When a ductile component (e.g. a steel brace or damper in a building) is deformed plastically, **work is done** to rearrange its structure permanently. The energy goes into its **internal energy** (it warms up) and is **not returned** to the oscillation as kinetic or elastic energy.
- So energy is **removed from the oscillating system** and the **amplitude falls**.

- An elastic material, by contrast, stores the energy as elastic PE and gives most of it back, so it doesn't damp much.
- Any damping (air resistance, viscous oil dampers, rubber bearings) works the same way: work done against resistive forces removes energy from the oscillation.

153 Earthquake engineering

- Buildings use **ductile** steel frames and connections, which bend permanently and absorb energy in a severe earthquake rather than snapping (as a **brittle** material such as unreinforced concrete or cast iron would).
- Other dampers: oil-filled hydraulic dampers, rubber base isolators, tuned mass dampers at the top of tall buildings.
- Related energy-absorbing ideas: car crumple zones, climbing ropes that stretch and absorb the energy of a fall.

MARK-SCHEME LANGUAGE

- "Ductile materials deform **plastically**", so "energy is dissipated / work is done" in permanent deformation. A ductile material is one that shows "large plastic deformation before it breaks"; it is not "stiff" or "elastic" (WPH15 Jun 2023, Jan 2026).
- Steel is ductile (WPH15 Oct 2024).

Often asked as: MCQs only so far (which describes a ductile material; why ductile materials reduce amplitude), set in the context of buildings in earthquake zones.

COMMON MISTAKE

Confusing ductile with brittle, or with elastic. Saying the energy is "lost" without saying it is transferred out of the oscillating system (to internal energy of the material and the surroundings).

Topic 5.6 – Astrophysics and Cosmology

SPEC 154 Gravitational fields

WHAT THIS SPEC POINT ASKS

Understand that a gravitational field (a force field) is a region in which a mass experiences a force.

154 What a field is

- A **gravitational field** is a **region where a mass experiences a force**. Every mass creates one, and gravity is **always attractive**.
- **Field lines** show the direction of the force on a small test mass placed there. The closer together the lines, the stronger the field.
- **Radial field**: around a spherical mass (a planet, a star) the lines point **inwards towards the centre**, and spread out with distance, so the field gets weaker. Outside the sphere it is the same as if all the mass were at the centre (a point mass).
- **Uniform field**: close to the Earth's surface, over a region small compared with the Earth, the lines are almost **parallel and equally spaced**, so g is effectively constant (9.81 N kg^{-1}). That is why mgΔh works for small height changes but not for satellites.

MARK-SCHEME LANGUAGE

- "A region where a mass experiences a force."
- Comparison with electric fields: "electric fields exert forces on charges whereas gravitational fields exert forces on masses" (WPH15 Oct 2020).

Often asked as: a short definition; MCQs comparing gravitational and electric fields (both radial around a point source).

COMMON MISTAKE

Defining the field as "where an object feels gravity": say **mass** and **force**. Drawing radial field lines pointing outwards.

SPEC 155 Gravitational field strength g = F/m

WHAT THIS SPEC POINT ASKS

Understand that gravitational field strength is the force per unit mass, g = F/m, and use this equation.

155 Definition

$$g = F/m$$

- g = gravitational field strength (N kg^{-1}); F = gravitational force on the mass (N); m = mass (kg).
- g is the **force per unit mass** on a small test mass. It is a **vector**, pointing towards the mass causing the field.
- N kg^{-1} is the same as m s^{-2}: g is also the acceleration of free fall at that point.
- **Weight** = mg. Your mass is the same everywhere; your weight depends on the local g.

155 Weightlessness is a misconception

- An astronaut in orbit is still in a strong gravitational field (about 8.7 N kg^{-1} at the height of the International Space Station). They **have weight**.
- They feel weightless because they and the spacecraft are **in free fall together**: both accelerate towards the Earth at the same rate, so there is **no contact force** between astronaut and floor. Gravity provides the centripetal force that keeps them in orbit (160).

WORKED EXAMPLE: WEIGHT ON MARS

g on the surface of Mars is 3.7 N kg^{-1}. Find the weight of a 70 kg astronaut there.
F = mg = 70 × 3.7 = **260 N** (on Earth: 70 × 9.81 = 690 N)

MARK-SCHEME LANGUAGE

- "Force per unit mass." Accept N kg^{-1} or m s^{-2} as the unit (WPH15 Oct 2025).
- Astronauts in orbit "are in a gravitational field so have weight", and "orbiting means there is a resultant (centripetal) force"; the statement that they are weightless "is not valid" (WPH15 Jan 2026).

Often asked as: calculate g on a planet or moon; explain why astronauts float; MCQ on where weight is zero (only at an infinite distance).

COMMON MISTAKE

Saying g is negligible at orbital height, so astronauts float. Using the planet's radius instead of the distance from its centre when the object is above the surface.

SPEC 156 Newton's law of gravitation

WHAT THIS SPEC POINT ASKS

Use $F = Gm_1m_2/r^2$ (Newton's law of universal gravitation).

156 The law

$$F = Gm_1m_2/r^2$$

- F = the attractive force between two point masses (N); G = 6.67 × 10^{-11} N m^2 kg^{-2}; m_1, m_2 = masses (kg); r = distance between their **centres** (m).
- Every mass attracts every other mass. The force on each body is the **same size** (Newton's third law), even if one is far heavier.
- Spheres act as point masses at their centres. For a satellite at height h above a planet of radius R: **r = R + h**.
- It is an **inverse-square law**: double r and F falls to ¼; triple r and F falls to ⅑.

WORKED EXAMPLE: EARTH–MOON FORCE

M_E = 5.97 × 10^{24} kg, M_{Moon} = 7.35 × 10^{22} kg, centre-to-centre distance 3.84 × 10^8 m.
$F = Gm_1m_2/r^2$ = 6.67 × 10^{-11} × 5.97 × 10^{24} × 7.35 × 10^{22} / (3.84 × 10^8)^2 = **1.98 × 10^{20} N**
The Earth pulls the Moon, and the Moon pulls the Earth, with this same force.

EXAM TIP

Ratio questions are quickest by scaling: if r goes up by a factor n, F goes down by n^2. When the answer is close to a claim, say so in numbers: "27.7, which is approximately equal to 28, so the claim is correct" scored; "27.7, so the claim is correct" didn't (WPH15 Jan 2025).

MARK-SCHEME LANGUAGE

- "Use of $F = GMm/r^2$" with r from the **centre**; combining with $F = mg$ to show $g\ r^2$ is constant (WPH15 Oct 2020).

Often asked as: compare forces or fields at different distances; inverse-square ratios; force between galaxies; MCQ force on a mass.

COMMON MISTAKE

Forgetting to square r. Using the height above the surface as r. Not converting km to m. Calculating the Sun's field at the Earth when its field at its own surface was asked for.

SPEC 157

Field of a point mass: $g = Gm/r^2$

WHAT THIS SPEC POINT ASKS

Derive and use $g = Gm/r^2$ for the gravitational field strength due to a point mass.

157 Derivation

- Put a small test mass m at distance r from a point mass M. Newton's law gives the force: $F = GMm/r^2$.
- Field strength is force per unit mass: $g = F/m = (GMm/r^2)/m$. The test mass **cancels**:

$$g = GM/r^2$$

- M = mass causing the field (kg); r = distance from its centre (m). So $g \propto 1/r^2$; GM is constant for a given planet, so $\mathbf{g\ r^2 = constant}$.
- At a planet's surface (radius R): $g_0 = GM/R^2$. Above the surface g falls: at $r = 2R$, $g = g_0/4$.

WORKED EXAMPLE: G AT THE HEIGHT OF THE ISS, AND THE EARTH'S MASS

(a) $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}, $M_E = 5.97 \times 10^{24}$ kg, $R_E = 6.37 \times 10^6$ m, ISS height 400 km.
$r = 6.37 \times 10^6 + 0.40 \times 10^6 = 6.77 \times 10^6$ m
$g = GM/r^2 = 6.67 \times 10^{-11} \times 5.97 \times 10^{24}/(6.77 \times 10^6)^2$ = **8.7 N kg^{-1}** (about 89% of its surface value)
(b) Mass of the Earth from g at the surface: $M = gR^2/G = 9.81 \times (6.37 \times 10^6)^2/6.67 \times 10^{-11}$ = **5.97×10^{24} kg**

MARK-SCHEME LANGUAGE

- Derivation: "use of $F = GMm/r^2$ with $F = mg$", "algebra to show $g = GM/r^2$", "statement that GM is constant" (WPH15 Oct 2020).

Often asked as: show from a graph that $g\ r^2$ is constant (read two or three pairs and calculate $g\ r^2$ each time); compare g on two planets; find a planet's radius from M and g.

BEYOND THE SPEC, BUT EXAMINED

- **Ratio method** for another planet: $g_P/g_E = (M_P/M_E) \times (R_E/R_P)^2$ (WPH15 Jan 2021).
- If you know a planet's density ρ: $M = \rho \times \frac{4}{3}\pi R^3$, so $g = \frac{4}{3}\pi G\rho R$ (WPH15 Jun 2021 MCQ).

SPEC 158

Gravitational potential in a radial field

WHAT THIS SPEC POINT ASKS

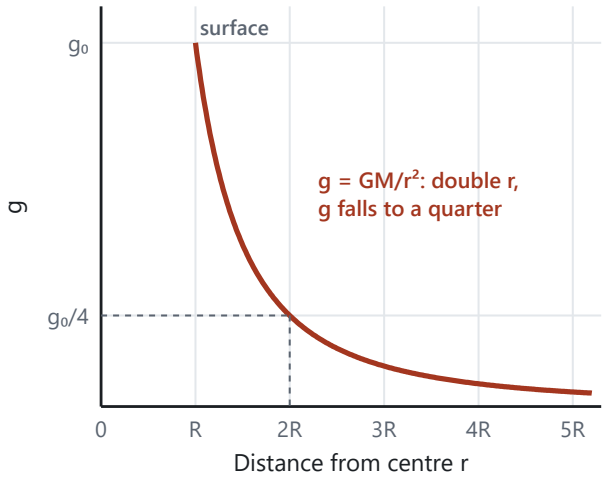
Use $V_{grav} = -Gm/r$ for the gravitational potential in a radial field.

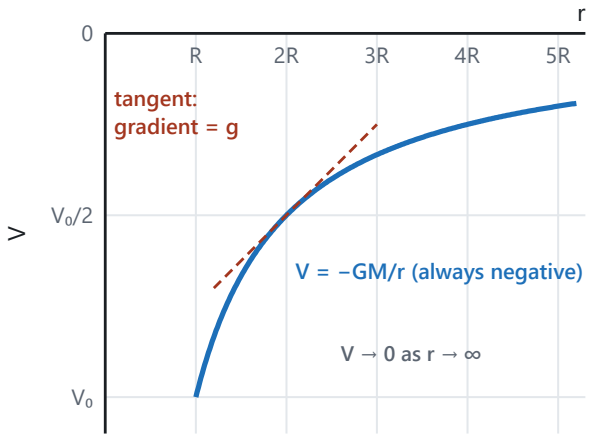
158 Definition

- The **gravitational potential** V at a point is the **work done per unit mass to bring a small mass from infinity to that point**. Unit: J kg^{-1}.
- Potential is **zero at infinity**. Gravity is attractive, so moving a mass **in** from infinity releases energy (work is done **by** the field). The potential is therefore **negative** everywhere at a finite distance.

$$V = -GM/r$$

- M = mass causing the field (kg); r = distance from its centre (m).
- V gets **less negative** (rises towards 0) as r increases. Its **magnitude** falls as 1/r (compare g, which falls as 1/r^2).





Outside a planet of radius R. Left: g = GM/r^2, so at 2R the field is a quarter of its surface value. Right: V = −GM/r, so at 2R the potential is half its surface value. The gradient of the V–r graph equals the field strength g.

158 Energy changes: ΔE = mΔV

- Gravitational potential energy of a mass m at r: $E_{grav} = mV = -GMm/r$.
- Moving between two points, the change in GPE is

$$\Delta E_{grav} = m\Delta V = m(V_{final} - V_{initial})$$

- Moving **away** from a planet, ΔV is positive, so energy must be supplied. Moving **towards** it, GPE falls and (with no other forces) KE rises by the same amount.
- **Don't use mgΔh for large height changes**: g isn't constant over thousands of kilometres.
- **Equipotentials** are surfaces of constant V: spheres around a planet, always perpendicular to the field lines. No work is done moving along one, which is why a circular orbit needs no energy to maintain.

158 Escape velocity

- To escape completely, an object must reach infinity (V = 0). Its KE at the surface must at least equal the energy needed: $\frac{1}{2}mv^2 = m \times (0 - (-GM/R)) = GMm/R$. So

$$v_{esc} = \sqrt{(2GM/R)}$$

- It doesn't depend on the mass of the object. For the Earth it is about 11 km s^{-1}.
- If the escape velocity at a body's surface would be greater than c, not even light can escape: a **black hole**. Setting $v_{esc} = c$ gives the radius $R = 2GM/c^2$.

WORKED EXAMPLE: LIFTING A SATELLITE INTO ORBIT

How much energy is needed to raise a 1200 kg satellite from the Earth's surface ($R = 6.37 \times 10^6$ m) to a height of 400 km? $GM_E = 6.67 \times 10^{-11} \times 5.97 \times 10^{24} = 3.98 \times 10^{14}$ N m^2 kg^{-1}.
V at surface = $-GM/R = -3.98 \times 10^{14}/6.37 \times 10^6 = -6.251 \times 10^7$ J kg^{-1}
V at orbit = $-GM/r = -3.98 \times 10^{14}/6.77 \times 10^6 = -5.882 \times 10^7$ J kg^{-1}
$\Delta V = -5.882 \times 10^7 - (-6.251 \times 10^7) = 3.69 \times 10^6$ J kg^{-1}
$\Delta E = m\Delta V = 1200 \times 3.69 \times 10^6 =$ **4.4×10^9 J**
($mg\Delta h$ would give 4.7×10^9 J: too big, because g is less than 9.81 N kg^{-1} above the surface.)

WORKED EXAMPLE: ESCAPE VELOCITY AND A BLACK HOLE

Escape velocity from the Earth: $v = \sqrt{(2GM/R)} = \sqrt{(2 \times 3.98 \times 10^{14}/6.37 \times 10^6)} =$ **1.1×10^4 m s^{-1}**
Radius a body with the Sun's mass (1.99×10^{30} kg) would need to be a black hole:
$R = 2GM/c^2 = 2 \times 6.67 \times 10^{-11} \times 1.99 \times 10^{30}/(3.00 \times 10^8)^2 =$ **2.9×10^3 m** (under 3 km)

MARK-SCHEME LANGUAGE

- "Use of $V = -GM/r$" at each radius, $r = R + h$; "use of $\Delta E = m\Delta V$".
- "Do not credit use of $\Delta E_{grav} = mg\Delta h$, as g is not constant" (WPH15 Jun 2021, Jan 2021, Jun 2025).
- "Do not award MP3 if radius of satellite is added to radius of Earth" (WPH15 Jun 2025).
- In $\Delta E = m\Delta V$, either mass can be the M in V, but the multiplier m must be the **other** mass (WPH15 Oct 2023).
- Escape-speed derivation: "GMm/r equated to $\frac{1}{2}mv^2$"; "do not credit $mg\Delta h$" (WPH15 Oct 2025).

Often asked as: potential at a height; energy to move a satellite between orbits; KE gained falling inwards; Earth's mass from a V against 1/r graph; derive and use escape velocity; black-hole radius.

COMMON MISTAKE

Using $mg\Delta h$ with g = 9.81 N kg^{-1}. Forgetting to add the height to the radius, or not converting km to m. Using g instead of G. Confusing potential (J kg^{-1}) with potential energy (J), or potential with field strength. Missing the unit. Saying V "increases" when you mean its magnitude.

BEYOND THE SPEC, BUT EXAMINED

- **Escape velocity** $v = \sqrt{(2GM/R)}$ and the black-hole radius (WPH15 Jan 2023, Oct 2025).
- A gas has a **range of molecular speeds**, so some molecules exceed the escape speed even when the r.m.s. speed doesn't, and the atmosphere slowly leaks away (WPH15 Jan 2023, Jan 2026).
- Reading M from a graph of V against 1/r (gradient = −GM) (WPH15 Jan 2025).

SPEC 159 Comparing gravitational and electric fields

WHAT THIS SPEC POINT ASKS

Compare electric fields with gravitational fields.

	Gravitational	Electric
Acts on	**mass**	**charge**
Field strength	$g = F/m$ (N kg^{-1})	$E = F/Q$ (N C^{-1})
Force between point sources	$F = Gm_1m_2/r^2$	$F = kQ_1Q_2/r^2$, $k = 1/4\pi\varepsilon_0$
Field of a point source	$g = GM/r^2$	$E = kQ/r^2$
Potential of a point source	$V = -GM/r$	$V = kQ/r$
Attract or repel?	**always attractive**	**like charges repel, unlike attract**
Sign of potential	always **negative**	positive near + charge, negative near − charge
Uniform field	near a planet's surface	between parallel charged plates
Strength	very weak	very strong

Similarities: both are **inverse-square** laws for point sources; both have **infinite range**; both give **radial** fields around a point/spherical source with spherical equipotentials; both have uniform fields with parallel, equally spaced lines; in both, potential $\propto 1/r$ and field strength $\propto 1/r^2$.

WORKED EXAMPLE: ELECTRIC V GRAVITATIONAL FORCE BETWEEN TWO PROTONS

$F_E/F_G = (ke^2/r^2)/(Gm_p^2/r^2) = ke^2/Gm_p^2$ (r cancels)
$= 8.99 \times 10^9 \times (1.60 \times 10^{-19})^2 / (6.67 \times 10^{-11} \times (1.67 \times 10^{-27})^2) =$ **1.2×10^{36}**
Gravity is negligible between particles; it dominates for planets and stars because bulk matter is electrically neutral overall.

MARK-SCHEME LANGUAGE

- Similarity: "both obey an inverse square law (for point masses / charges)" or "both have an infinite range".
- Difference: "electric fields can be attractive or repulsive, whereas gravitational fields can only be attractive", or "electric fields act on charges whereas gravitational fields act on masses" (WPH15 Oct 2020).

Often asked as: one similarity and one difference (2 marks); MCQs on radial fields and $1/r^2$ variation.

COMMON MISTAKE

Giving a "difference" that is really the same idea twice, or a similarity that is false (e.g. "both can repel").

SPEC 160 Orbital motion

WHAT THIS SPEC POINT ASKS

Apply Newton's laws of motion and his law of gravitation to orbits.

160 Gravity provides the centripetal force

- In a circular orbit the only force on a satellite is gravity, directed towards the centre of the planet. It is **perpendicular to the velocity**, so it changes the **direction** but not the **speed**. The resultant force is the centripetal force:

$$GMm/r^2 = mv^2/r = m\omega^2 r$$

- The satellite's mass m **cancels**, so the orbit doesn't depend on it.

160 Orbital speed and period

- From $GMm/r^2 = mv^2/r$:

$$v = \sqrt{(GM/r)}$$

- Using $\omega = 2\pi/T$ in $GMm/r^2 = m\omega^2 r$:

$$T^2 = (4\pi^2/GM)\, r^3$$

- Since G, M and π are constants, $\mathbf{T^2 \propto r^3}$ (Kepler's third law). A **lower** orbit is **faster** and has a **shorter period**.
- For two planets around the same star: $(T_1/T_2)^2 = (r_1/r_2)^3$.
- Shortcut: in orbit the centripetal acceleration equals the local field strength, $\omega^2 r = g$.

160 Geostationary orbits

- A **geostationary** satellite stays above the **same point on the equator**: it orbits **above the equator**, in the **same direction** as the Earth rotates, with **period 24 h**. Used for communications and TV (fixed dishes).
- A **polar orbit** passes over (or near) the poles in a lower orbit; as the Earth turns beneath it, it scans the whole surface. Lower height gives better resolution (weather, mapping), but it is only over a given place briefly.

WORKED EXAMPLE: GEOSTATIONARY HEIGHT

$T = 24\ h = 86\,400\ s$; $GM_E = 3.98 \times 10^{14}\ N\ m^2\ kg^{-2}$; $R_E = 6.37 \times 10^6\ m$.

$r^3 = GMT^2/4\pi^2 = 3.98 \times 10^{14} \times 86\,400^2/(4\pi^2) = 7.53 \times 10^{22}\ m^3$

$r = 4.22 \times 10^7\ m$

Height $= r - R_E = 4.22 \times 10^7 - 0.64 \times 10^7 =$ **3.6×10^7 m** (about 36 000 km)

WORKED EXAMPLE: PERIOD OF THE ISS

$r = 6.77 \times 10^6\ m$.

$T = 2\pi\sqrt{(r^3/GM)} = 2\pi\sqrt{((6.77 \times 10^6)^3/3.98 \times 10^{14})} = 5.55 \times 10^3\ s =$ **92 minutes**

$v = \sqrt{(GM/r)} = \sqrt{(3.98 \times 10^{14}/6.77 \times 10^6)} = 7.7 \times 10^3\ m\ s^{-1}$, about 16 orbits a day.

EXAM TIP

Set the equation up in words first: "gravitational force provides the centripetal force". Write $GMm/r^2 = m\omega^2 r$, then substitute $\omega = 2\pi/T$. Distances are always from the **centre** of the body being orbited.

MARK-SCHEME LANGUAGE

- "Equate $F = GMm/r^2$ with $F = m\omega^2 r$" (separate mark); "accept $F = mv^2/r$ with $v = 2\pi r/T$".
- Deriving $T^2 \propto r^3$: "statement that G, M (and π) are constants, so $T^2 \propto r^3$" (WPH15 Jun 2022: often missing).
- Lower orbit: "F is greater, and $F = m\omega^2 r$ means T must decrease", or "$T^2 = 4\pi^2 r^3/GM$ so $T \propto \sqrt{(r^3)}$" (WPH15 Jun 2021).
- Geostationary: "T = 24 h", $r = (GM/\omega^2)^{1/3}$, "height = r minus the radius of the Earth" (WPH15 Oct 2025).

Often asked as: orbital period or orbits per day compared with a claim; derive $T^2 \propto r^3$; period of a planet from the ratio method; geostationary height; why a lower orbit has a shorter period; effect of the Sun losing mass (smaller force, r increases, T increases).

COMMON MISTAKE

Adding the wrong radius (the satellite's or Pluto's) or forgetting to add the planet's radius. Not squaring 2π. Comparing a period in seconds with a claim in hours or days without converting. Writing an advantage when a disadvantage was asked.

BEYOND THE SPEC, BUT EXAMINED

- Energy of an orbiting satellite: KE = GMm/2r, GPE = −GMm/r, total = −GMm/2r. If its orbit drops, GPE falls and KE **rises**: it speeds up (WPH15 Jan 2024 MCQ).
- Air drag on a low satellite does work, raising the temperature of the satellite and the air around it (WPH15 Jan 2021).
- Geostationary v polar orbits: advantages and disadvantages (WPH15 Oct 2021). Time between two planets' alignments (synodic period) from their relative angular speed (WPH15 Jun 2022).

SPEC 161 Black-body radiators

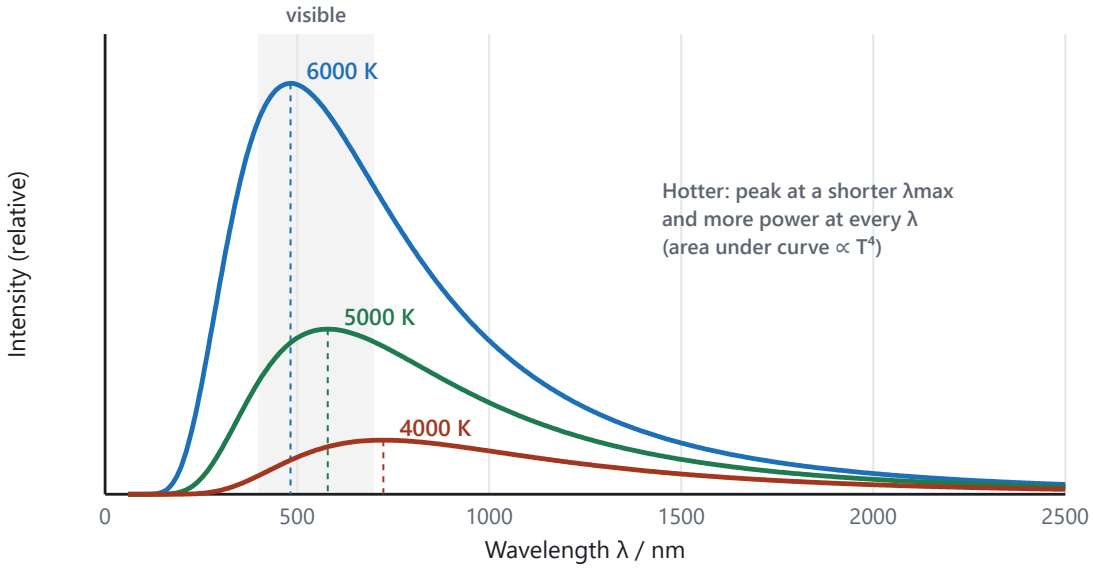
WHAT THIS SPEC POINT ASKS

Understand what a black-body radiator is, and interpret its radiation curves.

161 Definitions

- A **black body** is an idealised object that **absorbs all** the electromagnetic radiation falling on it (it reflects and transmits none). It is also the **best possible emitter** at any temperature.
- The radiation it emits depends **only on its temperature**. Everything above 0 K emits thermal radiation; at room temperature it is mostly infrared.
- Stars are a good approximation to black bodies. Real hot objects (a lamp filament, a glowing coal) are not perfect: they emit less than an ideal black body at the same temperature.

161 The curves



Black-body radiation curves. A hotter body has its peak (λ_{max}) at a shorter wavelength and emits more at every wavelength. The area under each curve is the total power, which rises steeply ($\propto T^4$).

Features to describe or draw:

- A **continuous** spectrum: some radiation at every wavelength. It rises steeply from short wavelengths to a peak, then falls more gradually.
- As temperature **rises**: the **peak moves to shorter wavelength** (Wien's law, 163); the **peak is higher**; the **total area** (total power) increases (Stefan–Boltzmann law, 162).
- Curves for different temperatures **never cross**: a hotter body emits more at every wavelength.
- Colour: a cooler star peaks in the red or infrared and looks red; a hotter one peaks in the blue or UV and looks blue-white. The Sun peaks near 500 nm (green), but because it emits across the whole visible range, it looks white.
- If the graph is plotted against **frequency** instead of wavelength, the hotter curve peaks at a **higher** frequency.

MARK-SCHEME LANGUAGE

- "λ_{max} decreases as the temperature rises"; "peak intensity and total power increase"; "the curve is continuous".
- A hot lamp filament is "not a perfect black-body radiator", or the glass bulb absorbs some radiation (WPH15 Jun 2025: fewer than 10% gave this; most wrote "energy is transferred to the surroundings").

Often asked as: MCQs comparing curves (including against frequency); "show that this body is not an ideal black body"; reading λ_{max} from a spectrum (163).

COMMON MISTAKE

Drawing curves that cross, or a hotter curve peaking at a longer wavelength. Thinking a black body must look black: a hot black body glows (the Sun).

SPEC 162 The Stefan-Boltzmann law

WHAT THIS SPEC POINT ASKS

Use $L = \sigma AT^4$ for a black-body radiator.

162 The law

$$L = \sigma A T^4$$

- L = **luminosity**: the total power emitted (W); σ = Stefan–Boltzmann constant = 5.67×10^{-8} W m^{-2} K^{-4}; A = surface area (m^2); T = surface temperature (K).
- For a spherical star of radius r: A = **$4\pi r^2$** (not in the booklet), so $L = 4\pi r^2 \sigma T^4$.
- $L \propto r^2 T^4$: temperature has a huge effect. Doubling T multiplies L by 16.

WORKED EXAMPLE: THE RADIUS OF THE SUN

The Sun has $L = 3.85 \times 10^{26}$ W and surface temperature 5800 K. Find its radius.
$r = \sqrt{(L/(4\pi\sigma T^4))} = \sqrt{(3.85 \times 10^{26}/(4\pi \times 5.67 \times 10^{-8} \times 5800^4))} = \sqrt{(4.78 \times 10^{17})} =$ **6.9×10^8 m**

WORKED EXAMPLE: RATIO OF LUMINOSITIES

Star X has twice the surface temperature and half the radius of star Y. Find L_X/L_Y.
$L_X/L_Y = (r_X/r_Y)^2 \times (T_X/T_Y)^4 = (½)^2 \times 2^4 =$ **4**

EXAM TIP

Many star questions chain three equations: **Wien** ($\lambda_{max} \rightarrow T$), **Stefan–Boltzmann** (T and r ↔ L), **intensity** (L and d ↔ I). Decide which quantity each step gives you before you start, and keep T in kelvin.

MARK-SCHEME LANGUAGE

- "Use of $L = \sigma AT^4$" and "use of $A = 4\pi r^2$" are separate marks (WPH15 Oct 2021).
- Comparison needs a unit: "candidates should provide a unit for L (or T), otherwise a valid comparison cannot be justified" (WPH15 Oct 2022 ER). L is in **W** (one mark scheme printed W m^{-2} in error).

Often asked as: radius of a star from L and T; T from L and r; L compared with a fraction of the Sun's; ratios of L or r; a probe or planet's energy balance.

COMMON MISTAKE

Using πr^2 or the volume of a sphere instead of $4\pi r^2$. Forgetting the fourth power. Using a star's distance d in place of its radius r. Not converting a wavelength to metres when Wien's law gives T first.

SPEC 163 Wien's law

WHAT THIS SPEC POINT ASKS

Use $\lambda_{max} T = 2.898 \times 10^{-3}$ m K for a black-body radiator.

163 The law

$$\lambda_{max} T = 2.898 \times 10^{-3} \text{ m K}$$

- λ_{max} = wavelength at which the intensity is greatest (m); T = surface temperature (K).
- $\lambda_{max} \propto 1/T$: hotter bodies peak at shorter wavelengths.
- The unit is **metre kelvin** (m K), not millikelvin: don't multiply by 10^{-3} as a "conversion".

WORKED EXAMPLE: TEMPERATURES FROM PEAKS

(a) The Sun's spectrum peaks at 500 nm. $T = 2.898 \times 10^{-3}/(500 \times 10^{-9}) =$ **5800 K**

(b) A red giant's spectrum peaks at 850 nm. $T = 2.898 \times 10^{-3}/(850 \times 10^{-9}) =$ **3400 K**

The red giant's peak is in the infrared, but it is still visible: it emits over a range of wavelengths, and the most intense **visible** light is at the red end, so it looks red.

MARK-SCHEME LANGUAGE

- "Accept a reasonable attempt to read λ_{max} from the graph" (WPH15 Jun 2026); show the reading on the graph, especially in a "show that".
- λ_{max} in the infrared "does not mean the star is invisible": there is a range of wavelengths about λ_{max}, and the visible part is mostly red (WPH15 Jan 2026, Oct 2021).
- Give T to at least 2 s.f. (WPH15 Jun 2023).

Often asked as: read λ_{max} from a spectrum and calculate T; peak wavelength of a star; colour of a star; red dwarfs on an HR diagram.

COMMON MISTAKE

Applying an extra 10^{-3} because of the "m K" unit. Not converting nm to m. Reading the peak of the wrong curve.

SPEC 164

Intensity and luminosity: the inverse square law

WHAT THIS SPEC POINT ASKS

Use $I = L/4\pi d^2$, where L is luminosity and d is the distance from the source.

164 Inverse-square law for radiation

- A star radiates power L equally in all directions. At distance d it is spread over a sphere of area $4\pi d^2$. The **intensity** (power per unit area received, W m^{-2}) is

$$I = L/4\pi d^2$$

- Intensity is what we measure at the Earth (apparent brightness). **Luminosity** is the star's total power output (an intrinsic property).
- Double the distance and the intensity falls to a quarter. This assumes nothing absorbs the radiation on the way (dust and gas would reduce I).

WORKED EXAMPLE: THE SUN'S LUMINOSITY FROM THE SOLAR CONSTANT

Just outside the Earth's atmosphere the Sun's intensity is 1360 W m^{-2}. The Earth–Sun distance is 1.50×10^{11} m.

$L = 4\pi d^2 I = 4\pi \times (1.50 \times 10^{11})^2 \times 1360 =$ **3.8×10^{26} W**

WORKED EXAMPLE: COMPARING TWO STARS

Star P is 3 times as luminous as star Q but twice as far away. How do their intensities at the Earth compare?

$I_P/I_Q = (L_P/L_Q) \times (d_Q/d_P)^2 = 3 \times (½)^2 =$ **0.75**: P appears slightly dimmer.

MARK-SCHEME LANGUAGE

- "Use of $I = L/4\pi d^2$". When you quote it in a written answer, **define I and L** (WPH15 Jan 2024 ER: symbols often undefined).
- Finish ratio or claim questions with a comparison (WPH15 Oct 2024: "Sirius appears 3 times, not twice, as bright").

Often asked as: L from I and d (Sun, a fast radio burst); d from I and L; ratio and claim questions; power collected by a detector or solar panel of known area ($P = IA$).

COMMON MISTAKE

Confusing intensity with luminosity (widespread). Using the radius of the star instead of the distance to it. Circular arguments (using $I = L/4\pi d^2$ to get L and then using the same equation to get d).

BEYOND THE SPEC, BUT EXAMINED

- Power received by a detector of area A: $P = IA$ (WPH15 Jun 2021).
- Energy balance on a planet: the planet intercepts sunlight over a disc (πR^2) but radiates from its whole surface ($4\pi R^2$), so the average intensity over its surface is a quarter of the incoming intensity (WPH15 Jun 2026).

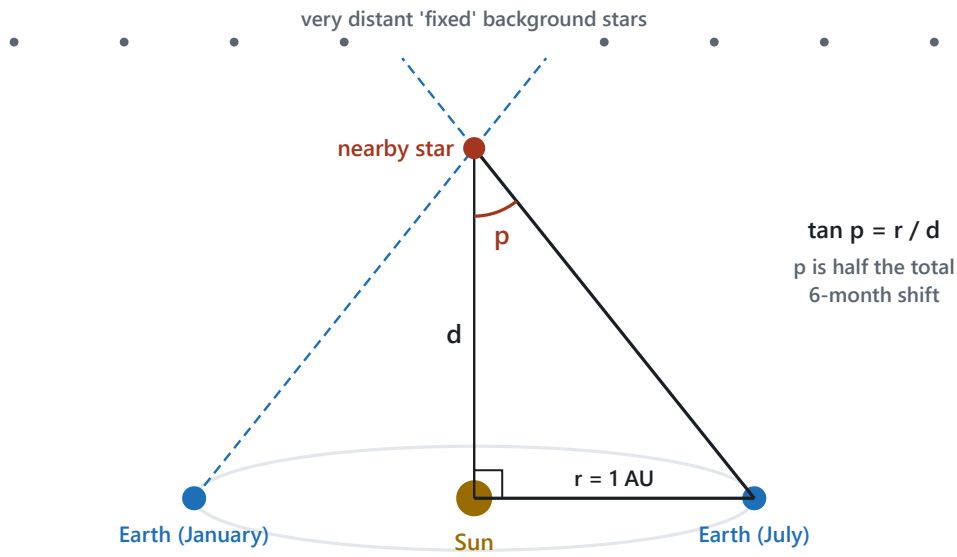
SPEC 165

Trigonometric parallax

WHAT THIS SPEC POINT ASKS

Understand how trigonometric parallax is used to measure astronomical distances.

165 The method



Trigonometric parallax. Viewed six months apart, from opposite ends of the Earth's orbit, a nearby star appears to shift against the very distant background stars. The parallax angle p is half the total angular shift; with r = 1 AU, tan p = r/d.

1. Measure the angular position of the nearby star **against the background of very distant ("fixed") stars**.
2. Measure it again **six months later**, when the Earth is on the **opposite side of its orbit**: the baseline is the diameter of the orbit.
3. The star appears to have shifted by an angle 2p. The **parallax angle p** is half of this.
4. Knowing the **radius of the Earth's orbit** (r = 1 AU = 1.50×10^{11} m), use trigonometry:

$$\tan p = r/d \quad \text{so} \quad d = r/\tan p \approx r/p \quad (p \text{ in radians, small})$$

165 Units

- 1 arcsecond (″) = 1/3600 of a degree.
- **Parsec** (pc): the distance at which the parallax angle is 1″. 1 pc = 3.09×10^{16} m = 3.26 light-years. Then **d (pc) = 1/p (arcsec)**.
- **Light-year**: the distance light travels in a year = $3.00 \times 10^{8} \times 3.15 \times 10^{7} = 9.46 \times 10^{15}$ m.

165 Limits

- The further the star, the **smaller** the parallax angle. Beyond a few hundred parsecs (from the Earth's surface; much further from satellites such as Gaia) the angle is too small to measure accurately: it is smaller than the telescope's resolution, and the **percentage uncertainty** becomes too large. Further stars need standard candles (166).

WORKED EXAMPLE: DISTANCE FROM PARALLAX

A star has a parallax angle of 0.20″. Find its distance in pc and in m.

d = 1/p = 1/0.20 = **5.0 pc**

In radians: p = 0.20/3600 × π/180 = 9.70×10^{-7} rad

d = r/tan p = 1.50×10^{11}/tan(9.70×10^{-7}) = **1.5×10^{17} m** (check: $5.0 \times 3.09 \times 10^{16}$ m)

MARK-SCHEME LANGUAGE

Describe the method (WPH15 Oct 2021, Jan 2023, Oct 2025):

- measure the "change in angle" / angular position of the star "over a 6-month period" (or across the diameter of the Earth's orbit);
- "against the background of distant (fixed) stars";
- "the radius / diameter of the Earth's orbit (about the Sun) must be known" and "trigonometry is used to calculate the distance" ("do not accept Pythagoras").
- A labelled diagram is accepted.
- Limit: "the parallax angle is too small (to be measured)", the uncertainty is too high: the idea that the **angle can't be measured**, not that the star can't be seen (WPH15 Jun 2025).

Often asked as: describe the method; show the method fails beyond some distance by comparing the angle with a telescope's resolution (allow sin θ or tan θ); MCQ on what parallax is for.

COMMON MISTAKE

Inverting the triangle (d = tan p/r). Forgetting to say the Earth–Sun distance must be known. Calculator in degrees when using radians (or vice versa). Using the full 6-month shift as p.

SPEC 166 Standard candles

WHAT THIS SPEC POINT ASKS

Understand how distances can be found from the intensity received from standard candles (objects of known luminosity).

166 The idea

- A **standard candle** is an **astronomical object of known luminosity**.
- Measure the **intensity** I received at the Earth. Since L is known, use the inverse-square law to find the distance:

$$d = \sqrt{(L/4\pi I)}$$

166 Examples

- **Cepheid variable stars**: their brightness varies regularly, and the **period** of the variation is linked to their **luminosity** (period–luminosity relationship: longer period, more luminous). Measure the period → read L from the relationship → measure I → find d. Useful to tens of megaparsecs.
- **Type Ia supernovae**: a white dwarf in a binary system that explodes always reaches nearly the **same peak luminosity**. So bright they can be seen in very distant galaxies (thousands of megaparsecs).

166 The distance ladder

- Parallax gives distances to nearby stars, including some Cepheids, which **calibrates** the Cepheid period–luminosity relationship. Cepheids in nearby galaxies that also host type Ia supernovae calibrate the supernovae. Each rung extends the range of the one before.
- Uncertainties: dust absorbs light (I measured too low, d too big); the calibration of L has its own uncertainty, which carries up the ladder.

WORKED EXAMPLE: DISTANCE TO A CEPHEID

A Cepheid's period gives its luminosity as 3.0×10^{30} W. The intensity received is 2.0×10^{-13} W m^{-2}. Find its distance.

$d = \sqrt{(L/4\pi I)} = \sqrt{(3.0 \times 10^{30}/(4\pi \times 2.0 \times 10^{-13}))} =$ **1.1×10^{21} m** (about 35 kpc, or 1.2×10^{5} light-years)

MARK-SCHEME LANGUAGE

- Standard candle: "a (stellar) object of **known luminosity**" (do not accept intensity or brightness for luminosity) (WPH15 Jun 2021, Jun 2025).
- Method: "measure the intensity received (on Earth)", "the luminosity is known", "use the inverse square law / $I = L/4\pi d^2$ (with symbols defined) to determine the distance". Intensity is "determined", not "calculated" (WPH15 Jun 2023, Jun 2025).
- Distance ladder (6-mark) (WPH15 Oct 2022): parallax to a nearby standard candle, measure its intensity and find L; find a standard candle of the same type in a distant galaxy; measure its intensity; use the inverse-square law.
- Cepheids: "measure the period of the intensity variation", use the period–luminosity relationship to find L.

Often asked as: define a standard candle; describe how one gives a distance; the distance-ladder 6-marker; Cepheid period from several cycles on a light curve; read a log period–luminosity graph.

COMMON MISTAKE

Mixing the standard-candle method with parallax in one answer. Confusing the standard candle with the galaxy it is in. Not defining I and L. Reading a period from one cycle instead of several. Taking a gradient on a log graph instead of reading values.

BEYOND THE SPEC, BUT EXAMINED

- Cepheid period–luminosity relationship and type Ia supernovae as the named standard candles (WPH15 Jan 2025, Oct 2025).
- Reading values from a period–luminosity graph with logarithmic scales (WPH15 Jan 2025).

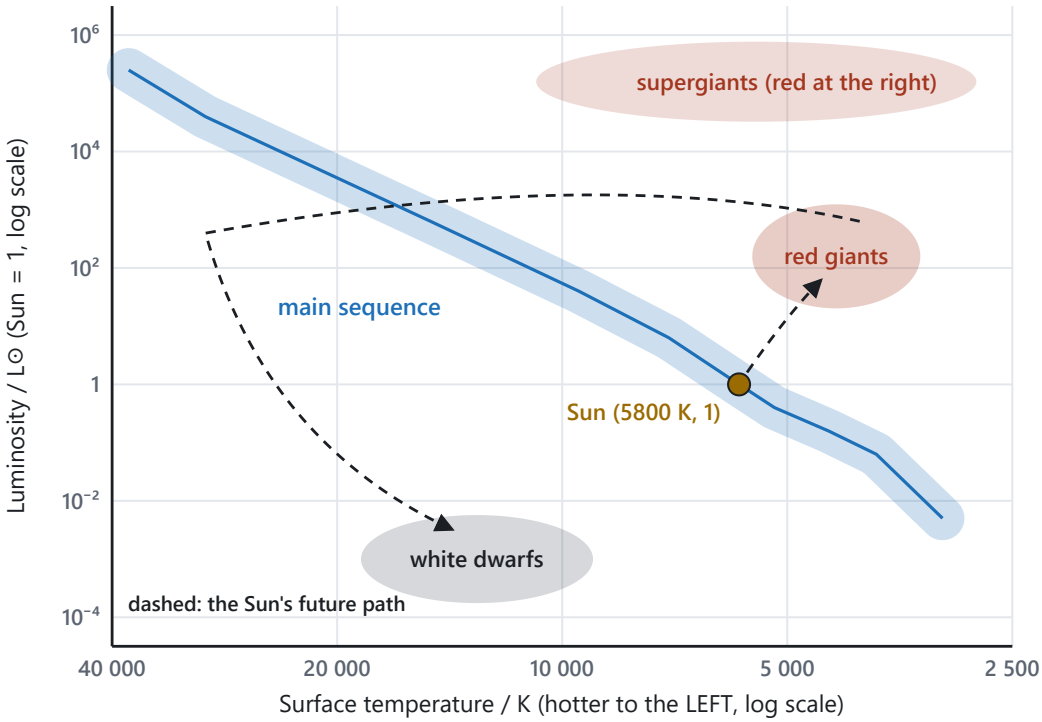
SPEC 167 The Hertzsprung-Russell diagram

WHAT THIS SPEC POINT ASKS

Sketch and interpret a simple Hertzsprung–Russell diagram of stellar luminosity against surface temperature.

167 The axes

- **y-axis**: **luminosity** (usually relative to the Sun, $L_\odot = 1$), on a **logarithmic** scale, typically 10^{-4} to 10^{6}.
- **x-axis**: **surface temperature** in K, on a **logarithmic** scale that runs **backwards**: **hottest on the left** (about 40 000 K) to coolest on the right (about 2500 K).



A schematic HR diagram. Most stars lie on the main sequence. Red giants and supergiants are cool but very luminous, so they must be huge; white dwarfs are hot but dim, so they must be tiny. The dashed line shows the Sun's expected path (168).

167 The regions

Region	Where	Temperature, luminosity	Why (from $L = 4\pi r^2\sigma T^4$)
Main sequence	diagonal band, top-left to bottom-right	hot and bright to cool and dim	stars fusing hydrogen in their cores; the more massive, the hotter and brighter
Red giants	above the main sequence, on the right	cool (3000–5000 K), L about 10–10^3	low T but high L, so a **large radius**
Red super-giants	top right	cool, L up to about 10^6	even larger
White dwarfs	bottom left	hot (about 10 000 K or more), L about 10^{-4}–10^{-2}	high T but low L, so a **tiny radius** (about Earth-sized)

- The **Sun**: about 5800 K and luminosity 1, in the middle of the main sequence.
- The diagram plots stars at one moment. A star's position changes as it evolves (168).

MARK-SCHEME LANGUAGE

- Axes: luminosity (log scale) up; temperature with a **reversed** scale, with units (K or °C, not "°K"), at least 2–3 values over a range like 2000–60 000 K (WPH15 Oct 2025, Jun 2026).
- Region names: "main sequence", "**red** giants", "**white** dwarfs"; "giants" or "dwarves" alone not accepted (WPH15 Jun 2023). "Red supergiants", not "supergiants" alone (WPH15 Oct 2025).
- Plot the Sun at luminosity 1 and about 5800 K (WPH15 Oct 2024).

Often asked as: label the axes with a reversed temperature scale; label regions A, B, C; mark the Sun; interpret a cluster's diagram (red giants and white dwarfs present, so not a young cluster).

COMMON MISTAKE

Temperature increasing to the right. Linear scales. Leaving the reversed temperature range blank. Writing "giants" without "red", or "dwarfs" without "white". Plotting the Sun in the wrong place.

BEYOND THE SPEC, BUT EXAMINED

- Reasoning about size from position, with $L = 4\pi r^2\sigma T^4$ (top right = huge, bottom left = tiny) (WPH15 Oct 2025).
- A cluster containing red giants and white dwarfs is old, because its massive stars have already left the main sequence (WPH15 Jun 2023).

SPEC 168

The HR diagram and the life cycle of stars

WHAT THIS SPEC POINT ASKS

Relate the Hertzsprung–Russell diagram to the life cycle of stars.

168 Birth and the main sequence

- A cloud of gas and dust (nebula) collapses under **gravity**, heating up into a protostar.
- When the core is hot and dense enough, **hydrogen fuses to helium**: the star joins the **main sequence**, where it spends about 90% of its life.
- On the main sequence the inward pull of **gravity** is balanced by the outward **pressure** from the hot gas and radiation. Position on the main sequence depends on **mass**: massive stars are hotter and brighter (top left).
- **Massive stars have shorter lives**: they have more fuel, but their much higher temperature makes the **rate of fusion** far higher, so they use up their core hydrogen much faster.

168 A Sun-like star (below about 8 solar masses)

1. **Hydrogen in the core runs out**: fusion in the core stops (the rate of fusion falls).
2. The core **contracts under gravity** and **heats up**. Hydrogen fuses in a shell around the core.
3. The outer layers **expand and cool**: the star becomes a **red giant** (moves up and to the right on the HR diagram: larger, cooler surface, more luminous).
4. The core **temperature rises enough for helium fusion** to start (helium → carbon and oxygen).
5. When fusion ends, the outer layers drift away (a planetary nebula). The hot core **collapses** to a **white dwarf**: small, very dense, hot, low luminosity, **no fusion**. On the HR diagram the star moves across to the left and down to the bottom-left corner.
6. The white dwarf slowly cools and dims (moves down and right).

168 A massive star (above about 8 solar masses)

- Becomes a **red supergiant**, fusing successively heavier elements, up to **iron**. Fusing iron releases no energy (iron has the highest binding energy per nucleon, 135).
- The iron core collapses suddenly; the star explodes as a **supernova**.
- The remaining core becomes a **neutron star**, or a **black hole** if it is massive enough (above about 3 solar masses).

MARK-SCHEME LANGUAGE

The six points in every version of the 6-mark (WPH15 Jan 2021, Jun 2024, Jun 2025, Jun 2026 (8 marks)):

- IC1 hydrogen is fused in the core (main sequence);
- IC2 when the hydrogen in the core is used up, the rate of fusion decreases / ceases;
- IC3 the core collapses / contracts under gravity;
- IC4 the temperature rises **to start** helium fusion ("not 'temperature rises after helium fusion starts'");
- IC5 the star expands into a **red giant** (larger area, lower surface temperature);
- IC6 when no further fusion takes place, the core / star collapses into a **white dwarf** (smaller, hotter).
- Massive stars "spend less time on the main sequence because of a higher rate of fusion" (WPH15 Jun 2025).

Often asked as: describe a Sun-like star's evolution using the letters marked on a given HR diagram (P, Q, S...); short versions worth 3 marks; why massive stars leave the main sequence first; properties of a white dwarf.

COMMON MISTAKE

Too much detail when the command word is "describe". Missing "red" or "white". Not using the letters on the diagram. Describing the balance of forces instead of the evolution. Bringing in fusion conditions (electrostatic repulsion), which answers a different question. Getting the order wrong: the core heats up **before** helium fusion starts.

SPEC 169

The Doppler effect

WHAT THIS SPEC POINT ASKS

Understand how relative motion between a wave source and an observer causes a shift in observed frequency.

169 Why it happens

- A source moving **towards** you: each wavefront is emitted a little closer to you than the last, so the wavefronts are **squashed together**. You receive a **shorter wavelength** and a **higher frequency**.
- A source moving **away**: the wavefronts are **stretched apart**: **longer wavelength**, **lower frequency**.
- The source itself emits the same frequency throughout; only the **received** frequency changes. The faster the relative motion, the bigger the shift. Only the component of velocity along the line of sight matters.
- Sound: a passing ambulance's siren drops in pitch as it goes by.

169 Light from space

- A star's spectrum has **absorption lines** at wavelengths fixed by the elements present. Comparing them with the same lines measured in a laboratory shows a shift:
  - **redshift**: lines shifted to **longer** wavelengths, so the source is **moving away**;
  - **blueshift**: lines shifted to **shorter** wavelengths, so the source is **approaching**.
- The whole pattern of lines shifts together, which is how you know it's the same element.

169 Uses

- **Rotation**: one edge of the Sun (or a galaxy) moves towards us and the other away. Light from the approaching edge is blueshifted, the receding edge redshifted. The difference between the two edges is

twice the shift of each.

- **Exoplanets**: an orbiting planet makes its star wobble. The star's velocity along our line of sight changes periodically, so its lines shift back and forth.
- **Binary stars** and the **expansion of the universe** (170).

MARK-SCHEME LANGUAGE

- "Light from the edge of the Sun moving towards the Earth is received with a **shorter wavelength**" (accept blueshift); "the receding edge gives a longer wavelength" (accept redshift) (WPH15 Oct 2021).
- "The source is moving away, so there is a Doppler shift and the wavelength received is greater" (WPH15 Oct 2024).
- Planet detection: the planet pulls the star, "so its velocity relative to the Earth varies", giving "a varying Doppler shift" (WPH15 Oct 2020).

Often asked as: explain a shift in context (Sun's rotation, a receding star, Andromeda approaching); planet detection; speed camera; MCQ (Doppler shifts can't directly show acceleration).

COMMON MISTAKE

Saying the lines "turn red": they shift towards longer wavelengths. Getting approaching and receding the wrong way round. Using the full edge-to-edge difference as one Doppler shift. Not stating the direction of motion.

BEYOND THE SPEC, BUT EXAMINED

- Edge-to-edge solar rotation: the measured difference is twice each edge's shift; many missed the factor of 2 (WPH15 Jan 2025).
- An exoplanet's tiny shift may be below the spectrometer's resolution (0.025 nm against 0.05 nm) (WPH15 Jun 2024).

SPEC 170 Redshift and Hubble's law

WHAT THIS SPEC POINT ASKS

Use $z = \Delta\lambda/\lambda \approx \Delta f/f \approx v/c$ for a source moving relative to an observer, and $v = H_0 d$ for objects at cosmological distances.

170 Redshift

$$z = \Delta\lambda/\lambda \approx \Delta f/f \approx v/c$$

- z = redshift (no unit); $\Delta\lambda = \lambda_{observed} - \lambda_{emitted}$; λ = the **emitted** (laboratory, rest) wavelength; v = speed of the source along the line of sight; $c = 3.00 \times 10^8$ m s^{-1}.
- Valid only when **v is much less than c**.
- Positive z (longer wavelength) means receding; a negative value means approaching (blueshift).
- $\Delta\lambda/\lambda$ is a ratio, so λ can stay in nm. Keep enough significant figures: the shift is usually a small difference between two close numbers.

170 Hubble's law

- Light from distant galaxies is **redshifted**, and the **further away** the galaxy, the **bigger** the redshift. So galaxies are receding with speeds proportional to their distance:

$$v = H_0 d$$

- v = recession speed; d = distance; H_0 = the **Hubble constant**, the gradient of a v–d graph.
- Units: with v in m s^{-1} and d in m, H_0 is in **s^{-1}**. Astronomers usually quote km s^{-1} Mpc^{-1} (about 70 km s^{-1} Mpc^{-1}). To convert, 1 Mpc = 3.09×10^{22} m:
  70 km s^{-1} Mpc^{-1} = $70 \times 10^3 / 3.09 \times 10^{22} = 2.3 \times 10^{-18}$ s^{-1}
- Every distant galaxy receding from every other is the evidence that the **universe is expanding**.

170 Age of the universe

- If the expansion rate has always been the same, everything was together a time t ago, where d = vt. With $v = H_0 d$:

$$t = d/v = 1/H_0$$

- With $H_0 = 2.3 \times 10^{-18}$ s^{-1}: $1/H_0 = 4.4 \times 10^{17}$ s ≈ 1.4×10^{10} years (14 billion years).
- A **smaller** H_0 gives an **older** universe (halve H_0 and the age doubles).

WORKED EXAMPLE: SPEED, DISTANCE AND AGE

A hydrogen line with rest wavelength 656.3 nm is observed at 662.9 nm in a galaxy's spectrum. H_0 = 70 km s^{-1} Mpc^{-1} (1 Mpc = 3.09×10^{22} m; 1 year = 3.15×10^7 s).
$z = \Delta\lambda/\lambda = (662.9 - 656.3)/656.3 = 6.6/656.3 = 0.0101$
$v = zc = 0.0101 \times 3.00 \times 10^8$ = **3.0×10^6 m s^{-1}**, receding (redshift)
$H_0 = 70 \times 10^3 / 3.09 \times 10^{22} = 2.27 \times 10^{-18}$ s^{-1}
$d = v/H_0 = 3.02 \times 10^6 / 2.27 \times 10^{-18}$ = **1.3×10^{24} m** (about 43 Mpc)
Age estimate: $1/H_0 = 1/2.27 \times 10^{-18} = 4.41 \times 10^{17}$ s = $4.41 \times 10^{17} / 3.15 \times 10^7$ = **1.4×10^{10} years**

MARK-SCHEME LANGUAGE

- "Must have the laboratory wavelength in the denominator" (WPH15 Oct 2024, Oct 2025); "allow use of $z = \Delta f/f$ and $z = v/c$".
- Approaching: a negative v, with the conclusion that the galaxy (e.g. Andromeda) "is moving towards the Earth" (WPH15 Jan 2021).
- Halving H_0 doubles the age; "H_0 is lower so the universe is older" alone gets only 1 mark (WPH15 Jun 2022).
- Testing Hubble's law from a plot: comment on a straight line through the origin, the scatter, and the gradient equal to H_0 (WPH15 Oct 2023).

Often asked as: speed from a wavelength or frequency shift; distance from $v = H_0 d$; convert H_0 to s^{-1} and find the age; H_0 from the gradient of a z–d graph (× c); order galaxies by redshift.

COMMON MISTAKE

Putting the observed wavelength in the denominator. Thinking Δλ is already given when the question gives two wavelengths. Rounding the wavelengths before subtracting. Mixing km and m in H_0. Forgetting "$\times 10^9$" when reading distances in billions of light-years. Saying $1/H_0$ gives the size of the universe (it gives an estimate of its age).

BEYOND THE SPEC, BUT EXAMINED

- H_0 in s^{-1} and in km s^{-1} Mpc^{-1}, with conversions given in the question; you don't need to know the megaparsec in advance (WPH15 Jun 2022, Jun 2025).
- $1/H_0$ assumes the universe has expanded at a constant rate (WPH15 Jun 2022).
- Light from very distant galaxies can be redshifted out of the visible into the infrared (WPH15 Oct 2022).

SPEC 171 The age and fate of the universe: H0 and dark matter

WHAT THIS SPEC POINT ASKS

Understand why the value of the Hubble constant and the possible existence of dark matter make the age and ultimate fate of the universe controversial.

171 The Big Bang

The universe began about 14 billion years ago in an extremely **hot, dense** state and has been **expanding and cooling** ever since. Space itself is expanding: galaxies are not flying through space away from a central point, and there is no centre. The evidence:

1. **Redshift of distant galaxies** (Hubble's law): everything is moving apart, so in the past it was closer together. Running the expansion back gives a starting time ($1/H_0$).
2. **Cosmic microwave background radiation (CMB)**: weak **microwave** radiation arriving from **every direction**, almost **uniform**, with a perfect **black-body spectrum** at about **2.7 K**. It was released about 380 000 years after the Big Bang, when the universe had cooled (to about 3000 K) enough for atoms to form and radiation could travel freely. The expansion since then has stretched its wavelength (redshifted it) about a thousand times, into the microwave region. A steady, unchanging universe can't explain it.
3. The observed amounts of **hydrogen and helium** (about 3:1 by mass) match what the hot early universe would have made.

171 Why the age is uncertain

- Age ≈ $1/H_0$, so the age is only as good as H_0. H_0 = gradient of v against d, and **distances to galaxies are hard to measure**: standard-candle luminosities are calibrated through the distance ladder, dust absorbs light, and each rung adds uncertainty.
- Different methods still disagree: measurements from the CMB give about 67 km s^{-1} Mpc^{-1} ($1/H_0$ ≈ 14.6 billion years), while Cepheids and supernovae give about 73 km s^{-1} Mpc^{-1} ($1/H_0$ ≈ 13.4 billion years). This disagreement (the "Hubble tension") is unresolved.
- $1/H_0$ also assumes a constant expansion rate. Gravity slows the expansion; observations suggest it is now speeding up. Either changes the true age.

171 The fate depends on density

Gravity from all the matter pulls back on the expansion. What happens depends on the **mean density** of the universe compared with a **critical density**:

Mean density	Universe	Fate
greater than critical	closed	expansion stops and reverses: everything contracts (a "Big Crunch")
equal to critical	flat	expansion slows, approaching zero but never quite stopping
less than critical	open	expands forever

The critical density is tiny, about 10^{-26} kg m^{-3} (a few hydrogen atoms per cubic metre), and it depends on H_0 ($\rho_c = 3H_0^2/8\pi G$), so an uncertain H_0 also blurs the dividing line.

171 Dark matter

- **Dark matter** is matter that has **mass** (so exerts a **gravitational force**) but **does not emit (or absorb) electromagnetic radiation**, so we can't see it.
- **Evidence: galaxy rotation curves.** For a star orbiting a galaxy's centre, gravity provides the centripetal force: $v = \sqrt{(GM/r)}$. If most mass were in the bright central bulge, stars far out would orbit more **slowly**. In fact the orbital speeds stay roughly **constant** far out (and orbital periods are shorter than predicted), so there must be much more mass, spread out, than we can see.
- Other evidence: **gravitational lensing** (light bent by more mass than is visible); galaxies in clusters moving too fast to stay bound by visible mass alone.
- Dark matter seems to be about **five times** as much as ordinary matter. Because we don't know exactly how much there is (or what it is), the **mean density is uncertain**, so we can't be sure whether the universe is open, flat or closed.
- Observations of distant type Ia supernovae suggest the expansion is **accelerating**. The cause is called **dark energy**, and it is not understood either.

WORKED EXAMPLE: EVIDENCE FOR DARK MATTER

A star orbits 5.0×10^{20} m from the centre of a galaxy whose visible mass (inside that orbit) is 1.0×10^{41} kg. It is observed to orbit at 2.2×10^5 m s^{-1}.
Predicted speed: $v = \sqrt{(GM/r)} = \sqrt{(6.67 \times 10^{-11} \times 1.0 \times 10^{41}/5.0 \times 10^{20})} = 1.2 \times 10^5$ m s^{-1}
The observed speed is almost double. Mass needed: $M = v^2r/G = (2.2 \times 10^5)^2 \times 5.0 \times 10^{20}/6.67 \times 10^{-11} =$ **3.6×10^{41} kg**
That is about 3.6 times the visible mass, so there must be a lot of mass that doesn't emit electromagnetic radiation: dark matter.

MARK-SCHEME LANGUAGE

- Dark matter "exerts a gravitational force" / "has mass" **and** "does not emit electromagnetic radiation"; "do not accept light for electromagnetic radiation" (WPH15 Jan 2026).
- Rotation data: "the observed period is smaller than expected, so the gravitational force is bigger, so there must be more mass than is visible" (WPH15 Jun 2024).
- Fate: open = density less than the critical density, expands forever; closed = greater, eventually contracts; flat = equal (WPH15 Oct 2024 MCQ).
- Dark matter adds to uncertainty about the fate because "the mean density (mass) of the universe is not known" (WPH15 Jan 2022).

Often asked as: MCQs on open, closed and flat universes and on what dark matter is; use a star's orbital period or speed to argue for dark matter; explain why the fate (or age) is uncertain.

COMMON MISTAKE

Answering a dark-matter question with redshift and Hubble's law. Saying dark matter must exist without using the data to show why. Saying it doesn't emit "light" (it must be electromagnetic radiation of any kind). Confusing "solar mass" with the mass of the dark matter. Saying the CMB is "left over from the explosion" without the cooling, the uniformity and the black-body spectrum.

BEYOND THE SPEC, BUT EXAMINED

- **Critical density** as the dividing line between open and closed universes (WPH15 Jun 2023, Oct 2024, Jan 2026 MCQs).
- Inferring dark matter from a star's orbital period or speed in a galaxy (WPH15 Jun 2024, Jan 2026).
- Uncertainty in H_0 (scatter on a Hubble plot) gives an uncertain age (WPH15 Jun 2022, Oct 2023).